

Soft Power Through (Humanitarian) Aid: Evidence from an RCT in Pakistan*

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Abstract

Can foreign aid shape recipients' values, attitudes, and political behaviours to advance donor interests? This question has motivated aid policy since the Marshall Plan and remains contentious as humanitarian needs reach unprecedented levels while donor budgets shrink. I provide experimental evidence through a randomised controlled trial with 2,450 vulnerable women in Pakistan, cross-randomising humanitarian aid receipt (four monthly payments of \$25) with information identifying the aid's foreign source. The interventions successfully address immediate needs and correct beliefs about aid origins. I find three key results for soft power—the ability to shape preferences through appeal rather than coercion. First, while aid and information each improve perceptions of foreign organisations when provided separately, their combination generates the strongest impact (10% increase). Second, the combination generates modest shifts in cultural tolerance, although broader value changes remain limited. Third, and most surprisingly, foreign aid information increases support for Pakistan's government by 0.1σ , due to recipients interpreting international presence as signalling improving domestic institutional prospects. These results demonstrate humanitarian aid's potential as a soft power tool, particularly when recipients understand its source.

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1 Introduction

Since the Marshall Plan, donors have pursued dual objectives through foreign aid: Addressing moral imperatives to help those in need while advancing strategic national interests by shaping recipients' hearts, minds and votes. This fundamental tension – whether aid can simultaneously serve humanitarian goals and function as “soft power,” the ability to shape preferences through appeal rather than coercion (Nye, 1990) – has shaped aid policy for decades. The question has become urgent as humanitarian needs reach unprecedented levels with 339 million people requiring assistance in 2024 (IRC, 2025), while donor nations slash budgets amid voter scepticism about the value of this aid for donor countries. The UK has cut aid to 0.3% of GNI, the US has dismantled USAID, and polls show 61% of Americans believe their government spends “too much” on foreign aid (Kaiser Foundation, 2013). The question of whether aid can advance national interests – long debated since Morgenthau (1962) framed it as an instrument of power – has become urgent in practice, as every dollar spent in aid needs to justify itself not only morally but strategically.

This tension is particularly acute for humanitarian aid, which comprises a substantial and growing share of total foreign aid flows. These transfers – reaching hundreds of millions of people and totalling \$46.9 billion in 2024 – provide immediate, life-saving assistance to populations facing crises from conflict, disasters, and displacement, usually in the most fragile of settings. While its moral objective is clear, humanitarian aid faces the deepest scepticism about strategic value. Donors have repeatedly attempted to leverage humanitarian assistance for geopolitical purposes, yet rigorous evidence on whether such attempts succeed remains lacking. For humanitarian aid to serve as soft power, it must do more than save lives: It must shape how recipients think about foreign actors, influence their fundamental values, and potentially affect their political behaviours.

In this paper, I study whether humanitarian aid can serve as an instrument of soft power, by shaping recipients' values, attitudes and political behaviours. I provide experimental evidence through a randomised controlled trial with 2,450 extremely vulnerable women in Pakistan, cross-randomising two interventions in a 2×2 design. First, randomly selected participants receive four monthly humanitarian aid payments of ~25 USD each – assistance designed to meet immediate basic needs. Second, I randomly provide information identifying the aid's source as an international donor, explicitly stating that the Pakistani government played no role. This research design enables me to pinpoint what exactly matters for soft power: The aid itself, the fact that it comes from a foreign donor, or the combination of both. This approach assesses whether humanitarian assistance can simultaneously achieve both strategic and humanitarian goals. I measure outcomes across three dimensions that are central to soft power and donors' goals: Perceptions of foreign actors, adoption of values donors promote, and political attitudes and behaviours.

Before examining soft power outcomes, I verify that both interventions achieved their intended objectives. Recipients spent aid primarily on food (93%) and medicine (76%), with corresponding improvements in humanitarian outcomes such as nutritional diversity, food security, access to medicine, and mental wellbeing. The information leads to participants meaningfully updating their beliefs about the source of the funds: While initially only 36% of participants (correctly) believe the aid comes from an international donor, receiving the information during the surveys increases this by 44 percentage points, or 120% compared to the control mean. There is also an increase in how certain participants are of these beliefs. As expected, these effects are concentrated among participants who initially wrongly believed that the aid did not come from an international donor, and persist over time across survey rounds. Overall, these results show that the interventions succeeded in 1) addressing humanitarian needs, and 2) correcting participants' beliefs about the source of the funds.

I first look at how the intervention affects participants' views on foreigners. At a fundamental level, donors seek to improve their standing in the countries in which they operate. This is important because more positive views towards donors decrease support for anti-donor activities, provide diplomatic leverage, and lend legitimacy in case of future interventions or projects in the recipient country. I find that both the aid and the information, when considered in isolation, improve participants' views on the work of international organisations helping Pakistan in general, and for women like them in particular. The effect of the aid itself is entirely driven by participants who correctly believed the aid came from an international donor to begin with. Moreover, the combination of the aid and the information has the strongest effect: It improves views on the work of international organisations by around 10% of the control mean, and significantly more than either the aid or the information in isolation. These results demonstrate the potential of humanitarian aid in enhancing recipients' perceptions of foreigners, particularly when they understand its source – a key soft power objective

I then examine whether the interventions have shifted a variety of values that donors have long sought to foster through aid. Western donors, specifically, have promoted liberal democratic values that are expected to advance market development, human rights, and peace, while fostering ideological alignment. Examining outcomes that capture key elements of these values – including universalism and positive-sum thinking (Enke, Rodriguez-Padilla, and Zimmermann, 2022; Chinoy et al., 2025) – I find limited effects in certain outcomes. The combination of aid and information makes participants 5-7% more respectful toward individuals from different cultures, belief systems and social groups, though other values remain unchanged. While these types of fundamental values are generally very hard to shift, it is noteworthy that even short-term humanitarian interventions can (moderately) influence them.

Lastly, one of the key soft power goals behind foreign aid has been to influence political behaviours in recipient countries, from strengthening friendly governments in challenging places, to undermining those considered hostile to the donor's strategic

interests. Using self-reported measures of views towards the government, I find that the combination of the aid and the information improves participants' views of the national government by $\sim 0.1\sigma$. These results are corroborated by exploiting only within-subject variation and using incentivised, lab-in-the-field games (Acemoglu et al., 2020; Blair, Marty, and Roessler, 2022), which allays concerns about the self-reported nature of these measures. This finding appears counterintuitive: Why would telling participants that the aid came from a foreign donor *increase* support for the national government? I demonstrate it is neither credit misattribution (the information explicitly states the government played no role) nor general life improvement (aid alone shows no effect). Instead, the mechanism appears to be more subtle: This combination conveys a positive signal about Pakistan's institutional trajectory, which in turn improves participants' views of the government. Indeed, the combination of the aid and the information makes participants more positive about the future of Pakistan's political system, especially for women like themselves. This suggests that humanitarian aid can paradoxically strengthen support for domestic governments when recipients interpret the foreign presence as a positive signal about their country's future prospects, rather than as evidence of state failure.

I find no effects on costlier political behaviours: Neither political participation inside or outside the household, nor a costly political task (signing a petition asking for more women's rights) responds to any treatment combination. These null results were anticipated in my pre-analysis plan. The precision of these estimates rules out even modest effects, confirming that while humanitarian aid can shift attitudes and perceptions, translating these into costly political actions requires either longer interventions or fewer structural barriers.

Overall, the results suggest that humanitarian aid has the potential to serve as a tool for soft power, in particular when combined with information about the source of the funds. The intervention successfully improves perceptions of foreign actors, generates modest shifts in cultural attitudes, and – most surprisingly – strengthens support for the domestic government through a signalling mechanism. These results prove robust to pre-specified experimenter demand tests and multiple hypothesis corrections, with high lab game comprehension and a 96% response rate across two rounds of in-person surveys.

This paper makes three contributions. First, it provides experimental evidence on whether humanitarian aid can function as soft power by shaping recipients' values, attitudes, and political behaviours at the individual level. The political consequences of aid have generated extensive debate since Morgenthau (1962) framed assistance as an instrument of power. Yet existing evidence relies on aggregate country-level analyses (Faye and Niehaus, 2012; Wright, 2009; Ahmed, 2012), politicians claiming credit for foreign-funded projects (Guiteras and Mobarak, 2015; Cruz and Schneider, 2017), or hypothetical aid through survey vignettes (Blair and Roessler, 2021). Closest to this paper, Lyall, Zhou, and Imai (2020) study the effect of a vocational and cash transfer

program in Afghanistan, on support for combatant groups in an active conflict zone rather than soft power objectives. My RCT with 2,450 humanitarian aid recipients provides causal evidence on whether this assistance can advance donor interests through soft power channels. By measuring effects on values, attitudes toward foreign actors, and political behaviours, and by cross-randomising information about the aid's foreign source,¹ I demonstrate that humanitarian aid can function as soft power – finding that effects depend critically on recipients knowing the aid's foreign origin, with the combination generating improvements in foreign perceptions, modest value shifts, and surprisingly, enhanced government support through signaling mechanisms.

Second, I contribute to our understanding of how foreign intervention impacts state-citizen relationships in fragile states.² Recent literature demonstrates that improved state capacity or service delivery strengthens political engagement (Acemoglu et al., 2020; Montenbruck, 2023; Weigel, 2020; Weigel and Kabue Ngindu, 2023). This paper examines a different side of this question: How citizens respond when foreign actors, not their government, provide essential services. The theoretical effect is ambiguous – foreign provision might undermine state legitimacy by highlighting government failure, or might enhance it by alleviating material hardship. My design, which randomises both aid receipt and information about its foreign source, allows me to disentangle these competing effects, finding that humanitarian aid increases government support when recipients interpret international presence as validation of institutional viability. This finding is especially important in fragile states where humanitarian flows concentrate, state capacity is minimal, and small shifts in values or political attitudes can reshape long-term social and political equilibria. By extension, this paper also contributes to the literature studying how political attitudes are formed and shaped (see, e.g., Cantoni et al., 2017; Giuliano and Tabellini, 2024; Nunn and Wantchekon, 2011), focusing on the role of foreign intervention in shaping these.

Third, I contribute to the literature on political effects of cash transfers by examining humanitarian assistance – a distinct and understudied transfer type that differs fundamentally from existing research. While studies of government-provided CCTs find mixed effects on political participation (Labonne, 2013; De La O, 2013; Zucco Jr, 2013) and individually-targeted transfers show minimal political impact (Blattman,

¹The information intervention also contributes to a literature studying the branding of foreign aid (De Juan, Hofman, and Koos, 2023; Dietrich, Mahmud, and Winters, 2018). In this paper, the cross-randomisation of the aid and the information allows me to disentangle whether what matters is the aid itself, the fact that it is foreign-provided, or the combination of both. The results demonstrate the importance of this information in achieving soft power goals.

²Other papers have looked at different types of foreign intervention, mostly military intervention (see, e.g., Dell and Querubin, 2018; Berman, Shapiro, and Felter, 2011; Miguel and Roland, 2011). Focusing on foreign aid, Nunn and Qian (2014) and Crost, Felter, and Johnston (2014) look at whether US food and development aid increase conflict in recipient countries, while Faye and Niehaus (2012) look at how foreign aid flows affect elections in developing countries. In this paper, I focus on a specific type of foreign aid – humanitarian aid – and study its micro-level effects on soft power outcomes.

[Emeriau, and Fiala, 2018](#); [Broockman et al., 2024](#)),³ humanitarian transfers merit distinct theoretical and empirical treatment due to their unique features. Unlike poverty alleviation programs, humanitarian aid provides emergency relief to crisis-affected populations facing acute deprivation. Crucially, it is delivered by foreign organisations in fragile states where sovereignty is contested and foreign presence politically charged. The combination of extreme vulnerability, foreign provision, and crisis contexts creates distinct political dynamics compared to domestic social protection. My results – showing that foreign identity matters critically for political effects – confirm that humanitarian transfers cannot be understood through the lens of standard cash transfer programs. This distinction matters as humanitarian aid totals \$46 billion annually and concentrates in politically unstable regions where understanding its political consequences is essential.

The rest of the paper is organised as follows. Section 2 describes the context in which the intervention took place and the sample. Section 3 describes the experimental design. Section 4 presents the results of the aid and information provision interventions. Section 5 presents the main results on the soft power outcomes. Section 6 discusses robustness checks. Section 7 concludes.

2 Context & Sample

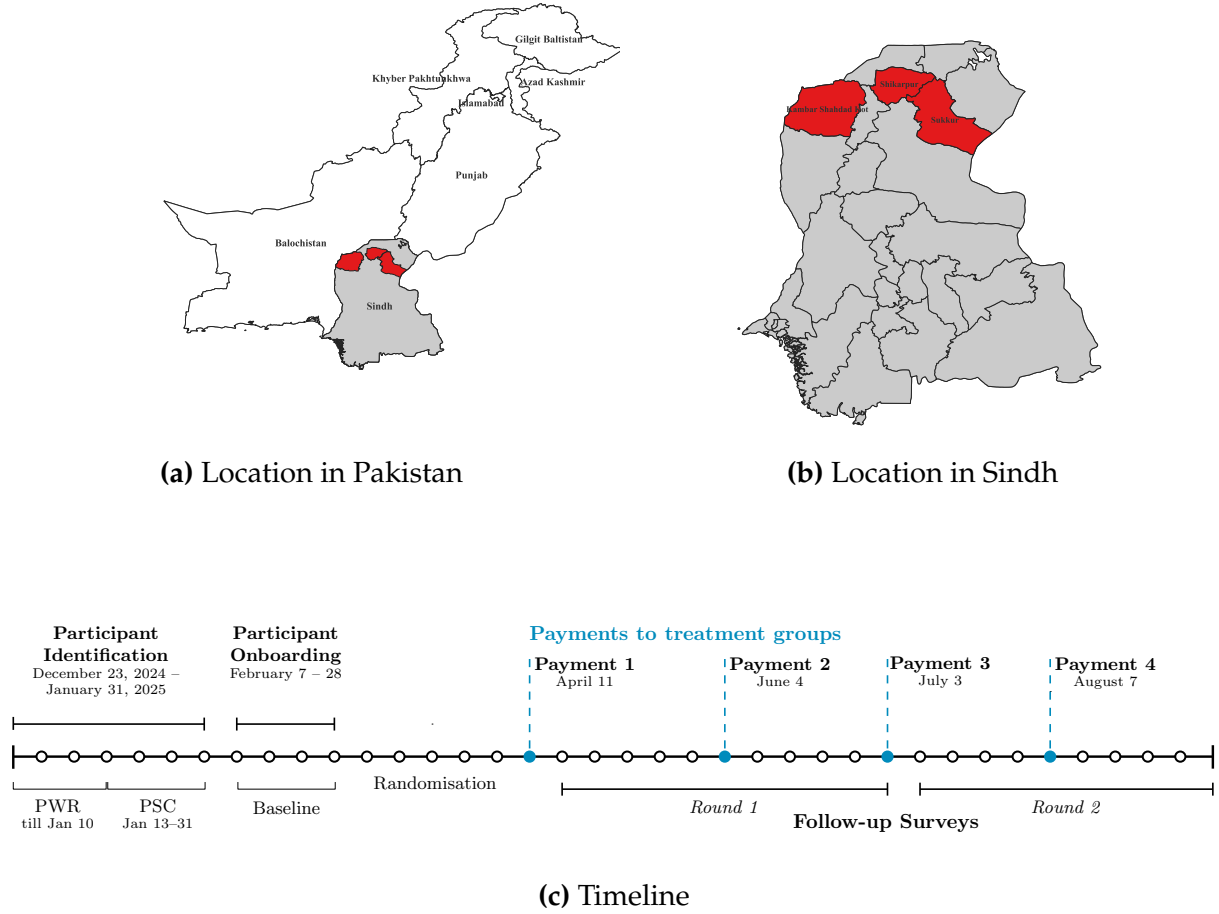
2.1 Location

Pakistan is an interesting setting to study this question. First, it has the fourth-largest population in need of humanitarian assistance ([Development Initiatives, 2023](#)). Given concurrent economic and climate crises, this is unlikely to change in the near future, making the questions studied in this project highly relevant for the local context. Second, Pakistan is undergoing an interesting political moment. The Prime Minister Imran Khan was removed from office in 2022, with the elections delayed until 2024 and widely criticised ([The Guardian, 2024](#); [BBC, 2024](#)), which has led to nationwide protests in recent times. The Economist’s Intelligence Unit categorised Pakistan in 2024 as authoritarian for the first time since 2006, and the OECD classifies it as a fragile country, with high fragility in the political dimension. Thus, small shifts in values, attitudes and political behaviours can have important consequences on Pakistan’s institutions.

The project took place in the main urban areas of three districts in Sindh, Pakistan: Kamber, Shikarpur and Sukkur (Figure 1). These districts were selected for two main reasons. First, our local partners operate only in Sindh. Second, these districts, while very poor, vary in their local conditions, providing interesting variation to exploit. In each district, we worked in 12 different communities.

³Yet more papers studying this question are [Manacorda, Miguel, and Vigorito \(2011\)](#), [Gromadzki, Sałach, and Brzeziński \(2024\)](#), [Hirvonen, Schafer, and Tukiainen \(2024\)](#) and [Zimmermann \(2021\)](#).

Figure 1: Locations & Timeline



Notes: The maps show the location of the three districts in which the intervention took place. Each node in the timeline denotes a week. Aid payments were made in weeks 16, 22, 27, and 32. The blue nodes highlight those payment weeks. PWR and PSC refer to the Participatory Wealth Ranking and Poverty Score Card exercises, respectively.

2.2 Participant Identification & Characteristics

Participants were selected through a rigorous two-step approach to identify the most vulnerable households in each community, which is explained in detail in [C](#) and briefly here (Figure 1 shows a timeline of the project). This was meant to have a sample with high levels of poverty and deprivation, similar to populations that receive humanitarian aid in other contexts. First, all households in the community were invited to a meeting in which the goal was to classify the community's households into four different socioeconomic groups ("Participatory Wealth Ranking", or PWR). To acknowledge local differences in what constitutes poverty/deprivation, participating households first identified a set of characteristics for each of the four socioeconomic groups and then classified each household into one of these groups. While moderators from our local partners were present to guide the activity, the ranking was based purely on the

discussion by participants, who were told the activity was to better understand the socioeconomic situation of the community, not to select participants for a humanitarian project, to avoid strategic answering.

The 150 households in the lowest socioeconomic group in each community were selected for the second step, which consisted of collecting a short questionnaire based on socioeconomic questions widely used in Pakistan for similar programs, plus some screening criteria ("Poverty Score Card", or PSC).⁴ Enumerators from our local partners visited each of the selected households to complete the questionnaire. The enumerators did not know the exact screening criteria nor the scoring rule based on the questionnaire to avoid manipulation. The responses were converted into poverty scores using a scoring rule, and the 70 poorest women (each representing a household) were selected for the study.

Among these 70 women in each community, 10 were set aside for the pure control group and 60 were invited to onboarding sessions. In these onboarding sessions, participants were informed for the first time that this was a humanitarian project and that two-thirds of participants would receive four aid payments worth 7,000 PKR (approximately 25 USD). Participants then provided their consent if they wanted to participate, and a short baseline survey was collected. Lastly, participants opened mobile wallets with the assistance of our mobile money partners, who also guided the women on how to use the wallets and provided them with a graphical brochure to aid their understanding. Our final sample consists of the 2,105 participants who attended the onboarding sessions and agreed to participate in the study (out of 2,160 invited) and the 360 in the pure control group, for a total of 2,465 participants.

Table 1 shows the baseline characteristics of the final sample. Participants are on average 40 years old, 84% have no education at all and 85% are married, with an average household size of 7.8 individuals. Around half state that they are the heads of their household. Participants are very poor, deprived and suffer from food insecurity: 97% are credit constrained (unable to raise a small amount of money in a short period of time), 22% and 50% are categorised as destitute and chronic poor, respectively, based on the poverty score, 53% and 33% have stressed or critical levels of food security (based on the Integrated Food Security Phase Classification, or IPC), respectively, and 47% and 46% are classified as being deprived and severely deprived based on the

⁴If there were more than 150 households in the lowest group, then a random sample of 150 was selected. If there weren't 150 households in the lowest group, then all households in the lowest group were selected, plus a random sample of households from the second lowest group needed to complete 150 households in a given community. In practice, almost all communities had enough households in the lowest group. In terms of screening, households were excluded if: They didn't have a valid national ID (CNIC) since they wouldn't be able to receive the aid payments; Were older than 65 years old or had a serious disability, since the piloting showed that these participants would struggle a lot to claim their funds and complete the surveys; or if they didn't have access to a mobile phone, which was needed to receive the payments (they didn't need to personally own a phone, but they did need to have access to a phone, for example from someone in the household). Fewer than 10% of participants were excluded for any of these reasons, see Table A5.

World Food Programme’s Multidimensional deprivation index (MDDI). Lastly, around three-quarters have experienced a shock in the previous 12 months (either man-made or natural), with low levels of mental well-being at baseline (less than one-fifth state that they are happy, and the average life satisfaction is 4.4 from 0 to 10). Overall, these characteristics suggest that the selection process successfully identified extremely poor and vulnerable participants.

Table 1: Baseline Characteristics

	Mean (1)	SE (2)	Percentiles					Obs. (8)
			10 th (3)	25 th (4)	50 th (5)	75 th (6)	90 th (7)	
Age	40.24	10.91	26	32	39	47	55	2465
Has no education	0.84	0.37						2465
Married	0.85	0.36						2350
Is head of household	0.52	0.5						2459
Involved in HH’s fin. decision-making	0.21	0.41						2350
Number household members	7.77	3.01	5	6	7	9	11	2351
Lowest socioecon. group (PWR)	0.76	0.43						2465
Income (last 30 days)	17212.73	9722.43	6000	10000	15000	22000	30000	2351
Receives BISP	0.46	0.5						2465
Credit constrained	0.97	0.18						2346
PSC: Destitute	0.22	0.42						2465
PSC: Chronic poor	0.5	0.5						2465
Stressed food security	0.53	0.5						2098
Critical food security	0.33	0.47						2098
MDDI: Deprived	0.47	0.5						2095
MDDI: Severely deprived	0.46	0.5						2095
Experienced shock (last 12 months)	0.74	0.44						2105
Happy (=1)	0.19	0.39						2347
Life satisfaction (0–10)	4.41	2.32	1	3	5	5	7	2351

Notes: Some variables were not collected for the pure control group, hence the reduced number of observations. For the pure control group, we use the values of the outcome variables (income, happy, and life satisfaction) at the time of their first survey.

3 Experimental Design

There are two main parts to this intervention. First, the intervention that randomly provides humanitarian aid to participants. Second, the intervention that randomly provides information on the source of the funds to participants. The goal of these two cross-randomised information was to be able to disentangle whether what matters is the aid itself, the fact that it comes from an international donor, or the combination of both. The experimental design is summarised in Figure 2.

Intervention 1: Aid Provision The main goal of the study was to understand the effects of humanitarian aid on a variety of outcomes, including soft power outcomes. To do so, participants were randomised into one of three arms, stratifying by their

Aid

Donor Information	No Aid	Receives Aid	
		Aid in Cash	Digital Aid
No Information			
Info on Donor Identity Only ----- + Values			

Figure 2: Experimental Design

community, their baseline needs and whether they receive assistance from Pakistan’s flagship social protection program (BISP):

1. *Control group (1065 participants)*: None of these participants received any aid payments and serve as a control group. Among them, 705 participated in the onboarding sessions and are thus aware of the humanitarian campaign (onboarding control group), while the remaining 360 did not participate in the onboarding sessions and thus are unaware of the humanitarian program (pure control group).

The two different control groups are used to study experimenter demand effects and strategic answering among participants, as explained in Section 6.1.

2. *Aid in Cash treatment group (699 participants)*: Participants in this treatment group received their aid through their national ID cards (CNIC), which they then needed to cash out. This is a “cash-light” way of delivering aid that mimics the most common way of delivering humanitarian aid nowadays.
3. *Digital Aid treatment group (701 participants)*: Participants in this treatment group received their aid through mobile wallets opened with and for them. This digital approach to delivering humanitarian aid is new and has many potential advantages over traditional aid delivery mechanisms. This approach has been shown to work in other similar contexts (see, e.g., [Callen et al., 2025](#)).

Participants in the treatment groups receive four monthly aid payments, each worth 7,000 PKR (around 25 USD), which is what our partners believe is necessary to cover the basic needs of the average family for a month. In a companion paper, [Fajardo-Steinhäuser and Findley \(2025\)](#), we study the relative effectiveness of the two different aid delivery mechanisms. However, for this study, I combine both treated arms into a single treated group, since the aid delivery channel is unlikely to play an important role in shaping the soft power outcomes of interest in this study.

Intervention 2: Information Provision Participants are not provided any information about the source of the funds at any stage. During the identification process, participants were told that these exercises were being conducted to understand the socio-economic situation of these communities. During the onboarding sessions, participants heard about the humanitarian program for the first time but did not receive any information about the source of the funds or the involvement of a foreign donor.

As shown in Figure 1, participants were surveyed twice: Once after receiving the first or the second payment, and again after receiving the third or the fourth payment. The payments were not linked to the surveys. During each survey, participants were asked questions on humanitarian outcomes, agency, and soft power outcomes, and two lab-in-the-field games were played.⁵ Each lab game was played twice, and the information was randomly provided immediately after participants had completed the first round of the games. The information provision was stratified by participants' treatment assignment in the aid intervention and whether they receive BISP. All soft power questions were asked after the information was provided. Thus, for the lab games, I can exploit within-individual variation from the rounds before and after the information is provided, while for the other outcomes, I can only exploit across-individual variation.

Two messages were provided to participants. Half of the participants ("Donor Identity Only") were told the following:

"As part of the project, 2 out of 3 participants that attended the event in February received payments of 7000 PKR to help them improve their economic situation. The money for this project and the payments came from an international donor outside of Pakistan. This international donor is an organization that funds projects related to economic development all over the world, including this. This international organization has funded other projects in Pakistan in the past. Without the funding of this international organization, it would not have been possible to conduct this project or to support the women in this project. The Pakistani government played no role in this project."

The goal of the message was twofold: First, to highlight that the money came from an international donor. Second, to explicitly state that the Pakistani government played no role in the project. The other half of participants received exactly the same information plus the following message ("Donor Identity plus Values"):

"This type of foreign assistance is much needed in Pakistan: 1 out of 5 people in Pakistan is undernourished, and one in three people living in districts like yours are facing serious levels of food insecurity. The work of international organisations like the one funding this project has prevented many deaths, sent many children to schools, and provided health services to people who would have otherwise not received these."

⁵Participants received a payment of 500 PKR for completing the survey, and up to 1,000 PKR based on their responses in the lab games. The average participant received a payment of ~1,350 per survey (around 5 USD).

The goal of this additional message was to highlight the values of the donor: Helping those most in need, despite the donor not being from Pakistan. In practice, the results are very similar across the two messages, so I only present the results combining those two groups together.

4 Impact of Interventions

In a first step, I confirm that the two interventions had their intended effect: The aid was spent on basic needs and led to improvements in humanitarian outcomes, and the information successfully corrected participants' beliefs about the source of the funding.

4.1 Intervention 1: Impact of Humanitarian Aid

*"Before this, we didn't have anything to eat.
At least now we can afford to have food,"*
34-year-old in Kamber

*"I bought groceries and medicines.
My child had been sick for a long time, so I got them treated,"*
23-year-old in Sukkur

*"We buy groceries, clothes for ourselves and our children,
we buy medicine for our kids that we were unable to buy before,"*
25-year-old in Shikarpur

The primary objective of a humanitarian program is to deliver immediate, short-term, and direct relief to the most vulnerable populations, helping them cover their basic needs. Thus, I look at whether participants spent their funds covering these basic needs and the aid led to improvements in humanitarian outcomes. For a deeper discussion of the humanitarian impacts of the aid, and the relative effectiveness of different aid delivery modalities, see [Fajardo-Steinhäuser and Findley \(2025\)](#), a companion paper.

Figure 3 shows how participants spent their funds. The two most common categories were food (93% of participants) and medicine (76%), supporting the claim that participants had very high levels of needs and vulnerability. This matches qualitative quotes from participants describing how the aid payments affected their lives (see above). Other common categories were transport (63%), debt repayment (39%), clothes (38%) and school items for their kids (34%). Very few participants used their payments to save (11%) or start a business (8%), which suggests that the main use was relaxing immediate constraints rather than escaping long-term poverty (which is beyond what humanitarian aid aims to accomplish).

These patterns match the impacts on humanitarian outcomes observed. Table 2 shows the impact of the humanitarian aid on four main humanitarian outcomes.

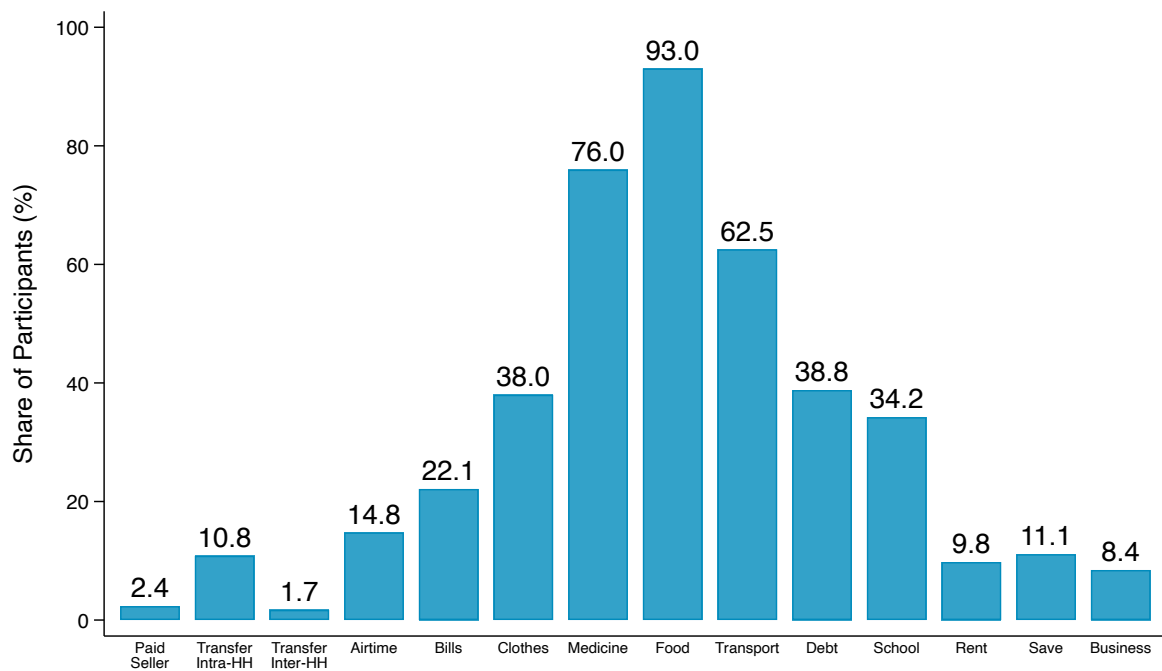


Figure 3: Aid Usage

Notes: The figure shows the proportion of participants who reported spending their aid money on different uses. Participants could indicate more than one use.

The first column shows that participants' nutritional diversity, as measured by their Food Consumption Score (FCS), significantly increases. This is driven by an increase in the consumption of meats and fruits, two items that were seldomly consumed by the households. The second column shows a significant decrease in food insecurity of 0.17σ , a large effect. This reduction is driven by participants skipping fewer meals and going fewer days without eating, with the impacts concentrated among children aged 0 to 4. The third column shows that the aid enabled participants to buy more medicine (conditional on having had a medical emergency in the last 30 days), with an effect of almost 50% the mean in the control group. Lastly, column 4 shows large improvements in a mental well-being index of 0.61σ (while this is very large, it is not unprecedented in humanitarian settings, see, e.g., [Callen et al., 2025](#)). Participants report being happier, more satisfied with their lives, and that their economic situation has improved.

Overall, these results suggest that the aid was used for its intended purposes (alleviating short-term needs) and significantly improved humanitarian outcomes among participants.

4.2 Intervention 2: Impact of Information Provision

At the beginning of each survey, participants were asked who they believed paid for the program. Enumerators were told not to read the options, but rather select the option closest to the participant's response. These beliefs were elicited at the end of

Table 2: Effect on Humanitarian Outcomes

	Nutritional Diversity (1)	Food Insecurity Index (2)	Afford to Buy Medicine (3)	Mental Wellbeing Index (4)
Received Aid	1.684*** (0.332)	-0.171*** (0.026)	0.129*** (0.046)	0.616*** (0.029)
Constant	34.338*** (0.258)	-0.001 (0.022)	0.289*** (0.011)	0.001 (0.022)
Observations	4,732	4,734	4,109	4,733
R ²	0.109	0.064	0.020	0.175

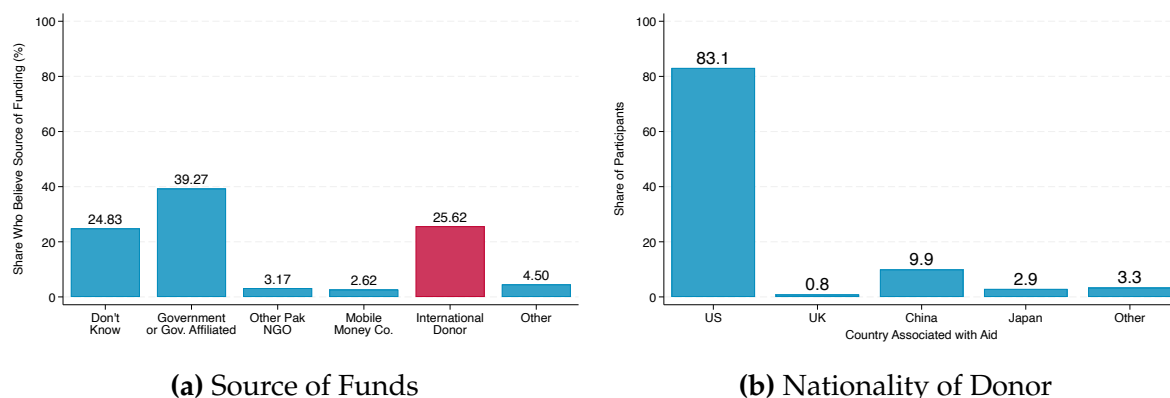
Notes: Robust standard errors in parentheses. Controlling for strata fixed effects and survey round fixed effects. Paid for medicine is conditional on having a medical emergency that required buying medicine.

the survey again to assess the effectiveness of the information intervention. Participants were also asked about their beliefs on the nationality of the donor in the second follow-up survey.⁶ Figure 4 shows the distributions of both beliefs. Panel A shows that most participants believe the aid comes from either the government or a government-affiliated organisation, with only a quarter of participants correctly identifying that the aid came from an international donor. Thus, there is a lot of room for these beliefs to be corrected. Panel B shows that most people associate any kind of foreign-funded humanitarian aid as coming from the US, with the second-most common nationality being China (less than 10%). Thus, even if participants are not explicitly told that the foreign aid came from an American donor (which was avoided to prevent any sensibilities associated with the US), most assume that is the case.

These results are important for several reasons. First, participants of development projects in general struggle to identify the funders of projects, despite the significant resources spent by governments, donor and humanitarian organisations in advertisement and propaganda (Dietrich, Mahmud, and Winters, 2018; De Juan, Hofman, and Koos, 2023). Second, it is likely that for (humanitarian) aid to work as a tool for soft power, recipients need to correctly identify that the aid came from an international donor or government. Lastly, participants misattributing the source of the aid could even lead to outcomes opposite of those intended by the funder, for example, the strengthening of an oppressive or adversarial government (Guiteras and Mobarak, 2015; Cruz and Schneider, 2017). The intervention here is very light-touch, simply stating that the money came from an international donor (which participants associate as being the US, see Panel B of Figure 4) and the Pakistani government played no role in the project.

⁶For those who correctly identified the donor to be foreign, they were asked for the identity of the donor. For the others, they were asked if the donor were foreign, what they believe is the most likely nationality of such a donor.

Figure 4: Respondents' Beliefs About Source of Aid and Nationality of Foreign Donor



Notes: Panel A shows the distribution of the beliefs participants had at the beginning of the first follow-up survey (before any information was provided) about the source of the aid. Panel B shows the beliefs of participants at the end of the second survey on the nationality of the donor, conditional on the donor being foreign.

Given that the beliefs on the source of the funds are elicited twice in each survey, once at the beginning and once at the end of the survey, I can measure the effectiveness of the information intervention in correcting recipients' beliefs about the source of the funds. Table 3 shows the results of this exercise. There are two outcomes of interest: Whether participants can correctly identify that the funds came from an international donor (odd columns) and their certainty in those beliefs (even columns). The Table includes data from only the first round of follow-up surveys, before any participant received the information. When looking at the full sample in the first two columns, receiving the information about the source of the funds significantly increases the probability of correctly identifying that the donor was a foreigner by 43.7 percentage points, an increase of 120% relative to the mean in the group that did not receive this information. Participants are also more confident in their beliefs, increasing their certainty by 0.86 points (on a scale from 0 to 10), more than 10% of the mean in the control group.

Exploiting participants' responses at the beginning of the survey, I can analyse whether the effects concentrate among participants who initially believed the money did not come from an international donor, as expected. This is shown in columns 3 and 4. For these participants, the ones whose beliefs needed correction, the effects are much larger: The information increases the correct identification of the source of the funds by 59.7 percentage points, a 400% increase relative to the control mean. Their certainty also increases by 1.18 points, or 15% of the control mean. When looking at participants who correctly identified the donor to be foreign at the beginning of the survey in columns 5 and 6, and for whom we would thus expect no effect from providing this information, we see that this is indeed the case: There is no effect from providing the information on either the correct identification of the donor or their certainty.

Table 3: Effect of Information Provision on Beliefs & Certainty

	Baseline Beliefs on Source					
	All Sample		Not Int. Donor		Int. Donor	
	Guessed Int. Donor (1)	Certainty (0-10) (2)	Guessed Int. Donor (3)	Certainty (0-10) (4)	Guessed Int. Donor (5)	Certainty (0-10) (6)
Received Information	0.437*** (0.021)	0.860*** (0.132)	0.597*** (0.021)	1.176*** (0.169)	-0.014 (0.018)	0.012 (0.131)
Constant	0.362*** (0.018)	8.209*** (0.118)	0.150*** (0.016)	7.783*** (0.152)	0.965*** (0.014)	9.396*** (0.099)
Observations	2,024	1,999	1,504	1,480	518	518
R ²	0.205	0.083	0.336	0.098	0.013	0.058

Notes: Robust standard errors in parentheses. Controlling for strata fixed effects. Odd columns show whether respondents correctly identify the identify of the funder (an international donor) at the end of the survey. Even columns show the certainty that respondents attach to their beliefs on the identify of the funder at the end of the survey. The first two columns look at the whole sample, the following two columns look at those who incorrectly believed at the beginning of the survey that the aid came from someone other than an international donor, and the last two columns look at those who correctly believed at the beginning of the survey that the aid came from an international donor.

These results show that the information provision intervention, despite being very light-touch, had a very large impact correcting participants' beliefs about the source of the funds, especially for those whose beliefs needed correction. Moreover, these effects persist over time: Table A1 presents the results using the answers to the questions at the beginning of the second round of surveys as outcomes. Despite an average gap of 79 days between the first and the second round of follow-up surveys, receiving the information in the first round increases the probability of correctly identifying the donor as foreign by 11.7 percentage points, or over 20% of the sample mean.

5 Humanitarian Aid as Soft Power

Having established that the interventions achieved their goals of 1) addressing participants' humanitarian needs and 2) correcting their beliefs about the source of the funds, I now turn to the main research question of whether humanitarian aid can be used as a tool for soft power, by changing recipients' values, attitudes and political behaviours. There are three main soft power outcomes of interest: Views on foreign actors, values and attitudes, and political behaviours.

Given the 2×2 cross-randomised design, and to be able to disentangle whether what matters is the aid itself, the fact that it comes from a foreign donor, or the combi-

nation of both, I estimate the following regression:

$$y_{it} = \beta_0 + \beta_1 \text{Control}_i \times \text{Donor Information}_{it} + \beta_2 \text{Aid}_i \times \text{No Information}_{it} + \beta_3 \text{Aid}_i \times \text{Donor Information}_{it} + \psi_{it} + \varepsilon_{it} \quad (1)$$

where y_{it} is the outcome of interest for participant i in survey round t . Control_i is a dummy that indicates whether the participant does not receive the humanitarian aid payments and is therefore part of the control group, while Aid_i indicates that the participant receives the humanitarian aid payments. $\text{No Information}_{it}$ is a dummy that indicates that the participant did not receive the information about the identity of the donor, while $\text{Donor Information}_{it}$ is a dummy that indicates that the participant received the information about the identity of the donor. ψ_{it} are strata and survey round fixed effects.⁷ I compute multiple-testing adjusted p -values following [Romano and Wolf \(2005\)](#), which are shown in square brackets. These are adjusted across the three relevant treatment arms, and for outcomes belonging to the same family and using the same sample, for consistency.

This analysis, along with the outcomes discussed below and the robustness checks, was pre-specified in a Pre-Analysis Plan registered at the AEA RCT Registry, trial number 0015121.

5.1 Views on Foreign Actors

Motivation & Measurement: At the most fundamental level, donors might want to use (humanitarian) aid as a way to improve their standing in the countries they operate. This has long been a goal behind soft power, and one of the most common concerns with the recent retreat of aid has been that it would be detrimental to donors' image. For example, a recent article in the [Council on Foreign Relations \(2025\)](#) argues that "An overzealous effort at reforming USAID will dismantle programs that enhance U.S. security, save lives, and boost United States' image around the globe." This is important because improving the views of the donor in recipient countries can lead to less support for anti-donor activities in those countries, diplomatic leverage, and increased legitimacy for the work and interventions of the donor.

I focus on one particular aspect of participants' views on foreign actors: Whether they believe that the work of international organisations is helping Pakistan as a whole, and women like them in particular. This maps closely to the information intervention, in which participants are told that the aid came from an international organisation.

Results: Table 4 shows the results for these outcomes. In the first two columns, looking at the whole sample, we see that the three arms have a significant and positive

⁷I am not controlling for baseline values of these outcomes since most of these outcomes were not collected at baseline, as some are considered sensitive and the baseline was taken in the presence of other participants.

effect on participants' views on the work of foreign organisations, both for Pakistan as a whole and for women like the participants. The effect for the former seems larger, although this difference is not statistically significant. Interestingly, this effect is present for those who receive the information but not the aid. This suggests that simply knowing that a foreign organisation is helping others in the community leads to improved views on the work of foreigners.

Table 4: Effect on Support for International Organisations

	All Sample		Believes Non. Int. Donor	
	<i>Agrees Work of Interntional Organisations is Helping...</i>			
	Pakistan	Women Like Her	Pakistan	Women Like Her
	(1)	(2)	(3)	(4)
β_1 : Control $\times$ Donor Info	0.033** (0.013) [0.011]	0.032** (0.014) [0.011]	0.008 (0.024) [0.964]	0.021 (0.025) [0.746]
β_2 : Received Aid $\times$ No Info	0.037*** (0.013) [0.006]	0.042*** (0.014) [0.004]	-0.009 (0.025) [0.964]	0.000 (0.027) [0.989]
β_3 : Received Aid $\times$ Donor Info	0.070*** (0.011) [0.001]	0.081*** (0.011) [0.001]	0.066*** (0.022) [0.008]	0.086*** (0.023) [0.001]
Constant	0.878*** (0.010)	0.872*** (0.010)	0.874*** (0.020)	0.860*** (0.021)
Observations	4,639	4,651	2,293	2,299
R^2	0.085	0.080	0.106	0.100
p -value $\beta_2 = \beta_3$	0.002	0	0	0

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses. Romano-Wolf multiple-testing adjusted p -values (Romano and Wolf, 2005), based on 1000 replications, are reported in square brackets, with adjustments applied jointly to Columns 1-2 and to Columns 3-4. Controlling for strata fixed effects and survey round fixed effects. At the bottom of the table, the p -value corresponds to a Wald test of the null hypothesis that the effect of receiving aid is the same with and without donor information. Columns 1 and 3 report results for the indicator of whether participants agree that the work of foreign organizations is helping Pakistan overall, while Columns 2 and 4 report results for the indicator of whether participants agree that the work is helping women like them. The first two columns use the full sample, whereas the last two restrict the sample to those who believed the aid was not from an international donor.

There is also an improvement in the views on foreigners for participants who receive the aid but not the information. This effect is entirely driven by those individuals who correctly identify that the donor is a foreign organisation, though. In columns 3 and 4, I focus on those who incorrectly believed the money came from someone other than an international donor. For those participants, receiving the aid only does not affect their views on foreign organisations.

Lastly, the effect is largest for those participants who receive both the aid and the information, since this combines two powerful signals: The fact that a foreign actor

provided the aid, and the positive impact that this aid has on participants' lives (as discussed in Section 4.1). The effect is significantly larger than for those receiving either the information or the aid alone. Moreover, this combination succeeds in improving views of foreign actors, even among participants who initially believed the aid did not come from a foreign donor. This shows that the combination of aid and information about the source of the aid can meaningfully improve recipients' perceptions of foreign actors, a key objective of soft power.

5.2 Values & Attitudes

Motivation & Measurement: Beyond improving perceptions of donor countries, soft power fundamentally seeks to shape the values and worldviews of recipient populations. For decades, Western donors have explicitly pursued value promotion through their aid programs, as evidenced by official policy documents and strategic goals. The [USAID Strategy on Democracy, Human Rights, and Governance \(2013\)](#) explicitly calls for promoting “human rights principles in accordance with universal values and international norms,” while the European Union frames its development cooperation as a means to “uphold and promote European values and interests” globally ([European Union, 2025](#)). This ideological dimension of aid represents a critical but understudied channel through which donors attempt to advance their interests – not merely by winning hearts and minds, but by fundamentally reshaping how recipients conceptualise fairness, cooperation, and social relations. This is particularly important since the global world order has been built upon the successful collaboration of people and nations with different cultures and principles, which can only be achieved if a certain alignment of values and ideologies exists.

While donor rhetoric often invokes broad concepts like “democratic values,” “market development” and “human dignity,” operationalising these abstract goals requires identifying specific, measurable dimensions of values that align with donor priorities. Market development, for instance, depends fundamentally on impersonal exchange – the ability to transact with strangers rather than relying solely on kinship networks. This requires populations to extend equal moral consideration to outsiders as they do to in-group members. Similarly, democratic participation requires citizens to view political competition as positive-sum rather than zero-sum, believing that one group's political gains need not come at the expense of others. Both market and democratic institutions also rely on generalised social trust and the willingness to cooperate beyond clan or ethnic boundaries – extending respect and recognition across social divisions. These specific moral orientations – universalism, positive-sum thinking, and cross-group respect – represent the foundational values that must shift before broader institutional changes can take hold. Influencing these underlying moral frameworks is key to achieving the goals of donors in fostering democratic governance or market-based development.

To capture these dimensions comprehensively, I measure seven distinct but related aspects of values and attitudes, drawing from recent advances in economics. Following [Enke, Rodriguez-Padilla, and Zimmermann \(2022\)](#); [Cappelen, Enke, and Tungodden \(2025\)](#), I assess universalism – whether people are equally altruistic and trusting toward strangers as they are toward in-group members. This captures whether individuals can move beyond their own group, which is essential for impersonal market transactions and inclusive democratic participation. I examine whether participants believe it is their duty to help others in their communities, particularly the poor, reflecting the social solidarity that supports redistributive policies in democratic societies. To capture the tolerance needed for diverse societies to function peacefully, I measure whether participants believe in respecting people from all cultures, religions, and social groups. I also assess positive-sum thinking following [Chinoy et al. \(2025\)](#) – whether individuals believe that when one group gets richer or gains political power, it does not come at the expense of other groups. Those who support this positive-sum worldview are more likely to support market competition and democratic alternation of power. Together, these measures provide a comprehensive assessment of whether humanitarian aid, particularly when its foreign source is made salient, can shift deeply held values in directions aligned with donor priorities – moving recipients from particularistic, kinship-based moral frameworks toward the universalistic principles that undergird functioning markets and democracies.

Results: Table 5 shows the results looking at these outcomes. A few patterns emerge. First, for most outcomes, none of the three treatments (information alone, aid alone, or the combination of both) seems to have a significant effect. This lack of significant effects is not due to a lack of statistical precision: To the contrary, these are precisely estimated null effects. For example, when looking at universalism or either measure of positive-sum thinking, the SEs on the coefficients would detect a significant effect at the 10% level of around 3% and 6% of the control mean, respectively, very small effects. The most likely explanation is that affecting something as fundamental as someone’s values requires a stronger, longer-lasting intervention. This intervention only lasted for four months (and half the surveys come from when participants had received only one or two aid payments), and the information provision is very light touch.

Second, and related to that point, the only significant effects come from the strongest treatment, receiving both the aid and the information. This combination has a positive and significant effect on participants’ respect towards individuals from other cultures and beliefs (column 4) or from other social groups (column 5). The effect sizes are small, however. Respect towards people from other cultures and beliefs increases by 5.2 percentage points, or ~7% of the control mean, while respect towards people from all social groups increases by 4.1 percentage points, or ~5% of the control mean.

Table 5: Effect on Values & Attitudes – All Sample

	Universalism (0-500)	Strong Moral Duty Help Poor	Responsible Others in Community (=1)	Respect All Cultures & Beliefs	Respect All Social Groups	Positive-Sum Thinking Ethnicity (=1)	Positive-Sum Thinking Income (=1)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
β_1 : Control $\times$ Donor Info	-3.558 (5.923) [0.993]	-0.008 (0.021) [0.993]	0.009 (0.012) [0.992]	-0.016 (0.020) [0.992]	-0.024 (0.018) [0.880]	-0.010 (0.022) [0.951]	0.002 (0.022) [0.993]
β_2 : Received Aid $\times$ No Info	4.192 (5.925) [0.993]	-0.008 (0.021) [0.993]	0.002 (0.012) [0.993]	0.024 (0.019) [0.922]	0.021 (0.017) [0.922]	-0.015 (0.022) [0.932]	0.012 (0.022) [0.951]
β_3 : Received Aid $\times$ Donor Info	-3.570 (5.069) [0.993]	0.012 (0.018) [0.993]	-0.004 (0.010) [0.993]	0.052*** (0.017) [0.022]	0.041*** (0.014) [0.055]	-0.001 (0.019) [0.993]	0.015 (0.019) [0.916]
Constant	283.764*** (4.012)	0.629*** (0.014)	0.083*** (0.008)	0.716*** (0.013)	0.801*** (0.012)	0.568*** (0.015)	0.557*** (0.015)
Observations	4,701	4,708	4,709	4,693	4,702	4,635	4,667
R^2	0.092	0.073	0.089	0.063	0.093	0.079	0.066
$\beta_2 = \beta_3$	0.145	0.291	0.566	0.118	0.182	0.467	0.891
$\beta_1 = \beta_3$	0.998	0.295	0.238	0.000	0.000	0.646	0.538

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses. Romano-Wolf multiple-testing adjusted p -values (Romano and Wolf, 2005), based on 1000 replications, are reported in square brackets, with adjustments applied to two groups: Columns 1-5 and Columns 6-7. Controlling for strata fixed effects and survey round fixed effects.

Column 1 reports allocations in the question, where the participant divides 500 PKR between a Pakistani stranger and a stranger from anywhere in the world (higher values indicate a greater allocation to the Pakistani stranger). Column 2 is an indicator for whether the participant agrees that the wealthy or strong have a moral duty to help the poor and weak. Column 3 is an indicator for whether the participant feels responsible for helping those outside their immediate family. Column 4 is an indicator for whether the participant believes in respecting all cultures and faiths. Column 5 is an indicator for whether the participant agrees that all social groups—men, women, rich, poor—deserve equal respect. Columns 6 and 7 report positive-sum thinking, namely whether the participant believe that when one group gains political power or gets richer, it does not come at the expense of other groups. Column 6 refers to the former, and Column 7 to the latter (equal to one if the participant disagrees with the zero-sum statement).

5.3 Political Behaviours

Motivation & Measurement: Perhaps the most ambitious aspect of soft power through aid is influencing political behaviours in recipient countries. Historical examples demonstrate this long-standing goal: Diplomatic pressure during the Marshall Plan era led to the exclusion of Communist parties from government coalitions in countries like France and Italy. More recently, a progress report shared with USAID and State Department officials highlighted how “public outreach is integrated into the design of each project to highlight the role of the Palestinian Authority in meeting citizens needs” with “events running every day... such that there is a constant stream of announcements and public outreach about positive happenings all over Palestinian areas in the critical week before the elections” (Faye and Niehaus, 2012). Such political influence, if achievable through humanitarian aid, would provide donors with a valuable geo-economic tool to shape outcomes in recipient countries – whether strengthening allied governments or catalysing change in hostile environments.

I comprehensively measure different types of political behaviours, reflecting the varying costs and constraints of political engagement for vulnerable women in Pakistan. At the most basic level, I use self-reported measures of government satisfaction, effectiveness, and legitimacy. Given the concerns regarding self-reported survey questions, I also use two lab-in-the-field games that provide incentivised measures of participants’ views on the government’s effectiveness and legitimacy (Acemoglu et al.,

2020; Blair, Marty, and Roessler, 2022). These attitudinal measures represent a key outcome of interest for soft power and the most malleable aspects of political opinion and perception.

I examine political participation, a costlier type of political outcomes, at two levels: within-household actions (such as asking family members to raise community issues with leaders) and outside-household engagement (including participation in women's groups addressing political concerns). This distinction is crucial given the severe mobility and autonomy constraints facing women in Pakistan, where even indirect political participation through male family members represents meaningful engagement. Finally, I include a real-world measure of costly political action: willingness to sign a public petition advocating for greater women's rights in the local area. Participants are informed that their names would appear publicly if they choose to sign, making this a credible signal of political commitment.⁸

This graduated approach – from private attitudes through household influence to public political action – allows me to identify precisely where the effects of humanitarian aid reach their limits. While aid might shift satisfaction or perceptions, influencing costly political behaviours represents a more stringent test of soft power's reach, particularly given the brief four-month intervention period and the profound constraints on women's political agency in this context, something that was highlighted in the pre-analysis plan.

Views on Government: The first set of outcomes concerns participants' perception of the government. Table 6 shows the results. The first five columns examine outcomes related to participants' attitudes toward the government, while the last three columns assess the legitimacy of various government components. It is important to note that government effectiveness/satisfaction and legitimacy, while related, are fundamentally different and capture distinct concepts. Effectiveness and satisfaction concern the government's ability to provide services, execute projects and steer the country in the right direction, while legitimacy concerns people's willingness to obey authorities and comply with government regulations and laws (Levi, Sacks, and Tyler, 2009).

In terms of attitudes, the results suggest that receiving the information alone does not affect these outcomes. Receiving the aid alone has a positive, but mostly insignificant, effect on these outcomes: The aid seems to increase participants' perceptions of government effectiveness in providing services, and the overall index combining these outcomes increases by 0.076σ (significant at the 10% level). The strongest effect comes from receiving both the information and the aid: This combination improves participants' perceptions of the government's effectiveness, increases their satisfaction with the government, and makes them more likely to believe that the government is go-

⁸In reality, their names were not made publicly available, nor was there a petition. This was included in the IRB applications.

ing in the right direction. Combined, there is an improvement in the index combining these outcomes of 0.125σ , a sizable effect. This sounds counterintuitive: Why would being told that a foreign donor provided assistance improve recipients' perceptions of the national government? While the mechanisms are discussed in detail below, this combination appears to convey a positive signal to participants about the country's future, which they attribute to the government.

Table 6: Effect on Political Attitudes & Legitimacy

	Attitudes					Legitimacy		
	Gov. Effective Providing Services (1)	Trust National Government (2)	Satisfaction with Nat. Gov. (3)	Gov. Going Right Direction (4)	Anderson Index (5)	Gov. Authority Make People Pay Taxes (6)	Police Right Make People Obey Law (7)	Courts Right Make Decisions People Abide to (8)
β_1 : Control $\times$ Donor Info	-0.022 (0.022) [0.743]	0.008 (0.022) [0.935]	0.029 (0.022) [0.614]	-0.005 (0.019) [0.950]	-0.001 (0.044) [0.990]	-0.006 (0.022) [1.000]	-0.012 (0.020) [0.987]	0.002 (0.018) [1.000]
β_2 : Received Aid $\times$ No Info	0.055** (0.022) [0.102]	0.019 (0.022) [0.751]	0.033 (0.022) [0.511]	0.021 (0.020) [0.743]	0.076* (0.044) [0.387]	0.025 (0.022) [0.879]	0.006 (0.020) [1.000]	-0.001 (0.018) [1.000]
β_3 : Received Aid $\times$ Donor Info	0.063*** (0.019) [0.007]	0.025 (0.018) [0.614]	0.061*** (0.018) [0.007]	0.041** (0.017) [0.114]	0.125*** (0.037) [0.007]	0.014 (0.019) [0.982]	0.003 (0.017) [1.000]	0.011 (0.015) [0.982]
Constant	0.519*** (0.015)	0.586*** (0.015)	0.575*** (0.015)	0.255*** (0.013)	0.005 (0.029)	0.426*** (0.015)	0.716*** (0.013)	0.803*** (0.012)
Observations	4,706	4,717	4,718	4,634	4,725	4,660	4,694	4,687
R^2	0.058	0.059	0.071	0.050	0.073	0.055	0.061	0.062
p -value $\beta_2 = \beta_3$	0.693	0.782	0.159	0.283	0.225	0.591	0.858	0.449

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses. Romano-Wolf multiple-testing adjusted p -values (Romano and Wolf, 2005), based on 1000 replications, are reported in square brackets, with adjustments applied jointly to Columns 1-5 and to Columns 6-8. Controlling for strata fixed effects and survey round fixed effects.

Column 1 is an indicator that is one if the participant finds the current national government effective in providing services to the average person. Column 2 is an indicator for whether the participant trusts the current national government. Column 3 is an indicator for whether the participant is satisfied with the current national government. Column 4 is an indicator for whether the participant believes the Government of Pakistan is going in the right direction. Column 5 is the Anderson (2008) index combining Columns 1-4. Column 6 is an indicator for whether the participant agrees that the government's tax authorities always have the right to make people pay taxes. Column 7 is an indicator for whether the participant agrees that the police always have the right to enforce the law. Column 8 is an indicator for whether the participant agrees that courts always have the right to make decisions that people are required to obey. Columns 1-5 present the effects on recipients' attitudes towards the government. Columns 6-8 present the effects on recipients' beliefs about the legitimacy of different government institutions.

In terms of legitimacy, participants' perceptions of the government's legitimacy to raise taxes, enforce the law through the police, and courts making decisions that citizens must abide by remain unchanged across the three groups. The combination of the aid and the information improves their perception of the government's effectiveness, but not its legitimacy.

Lab Games: Given the self-reported nature of the measures in the previous section, there are concerns that people might be answering in a non-truthful way, especially considering the sensitive nature of some of these questions. To overcome these concerns, I employ two lab-in-the-field games. One, used by [Acemoglu et al. \(2020\)](#) in Pakistan, aims at measuring government's effectiveness, while the other, used by [Blair, Marty, and Roessler \(2022\)](#) in Sierra Leone, aims at measuring government's legitimacy. Here, I provide a brief overview of the games' mechanics and their motivation, with a complete description of the games' protocols provided in Online Appendix E.

In the first game, the Government Fund Game, participants receive an endowment of 500 PKR (~2 USD). They then decide how much of this endowment they keep for themselves, and how much they give to a government fund. They are told that any money allocated to the government fund will be doubled and given to the government “to be used for local programs in your community”.⁹ Thus, participants will allocate more money to the government fund if they believe that the government will actually spend their funds in an effective manner in these local programs in their community.

In the second game, the Income Reporting Game, participants again receive an endowment of 500 PKR (“income”). They then need to decide how much of this income to report to the government, and how much not to report to the government. Any income reported to the government is taxed at a 25% rate, as would be the case with income taxation. If participants do not report part of their income, there is a 10% probability that they are audited. If they are audited, they receive a penalty of 50% of the income they did not report. Participants are told that the tax/penalty income collected in the game will actually be sent to the government.¹⁰ These values are not calibrated to match real-world rates but to make computations easier for participants. The idea here is that participants will report a higher share of their income if they believe the government to be legitimate, as they are complying with the rules of the government (paying taxes).

For each of the two games, participants play a practice round where surveyors walk them through the games in detail (participants’ understanding of the games, asked using four comprehension questions after the practice rounds, is very high, as seen in Table A2), and then they play two real rounds. For each game, they also play a non-political version of the game (for the Government Fund game, it is giving to a local sports club; for the Income Reporting game, it is reporting profits to an NGO). The information provision intervention takes place between the two real rounds of the games. This setup has two important advantages: First, the same games are played just before and immediately after the information is provided. Second, given the multiple rounds, it is possible to add individual fixed effects to the regressions, controlling for any individual-specific factors that remain constant across game rounds. Thus, for the games, I estimate the following regression:

$$y_{ir} = \gamma_0 + \gamma_1 \text{Received Aid}_i + \gamma_2 \text{Donor Information (Between Rounds)}_{ir} + \gamma_3 \text{Received Aid}_i \times \text{Donor Information (Between Rounds)}_{ir} + \omega_{ir} + v_{ir} \quad (2)$$

where y_{ir} is the outcome of interest for participant i in game round r . Received Aid_i indicates that the participant receives the humanitarian aid payments. Donor Informa-

⁹The endowment given to the government is doubled to give an incentive to participants to give to the government fund.

¹⁰In practice, implementing sending money from the games to the government was impossible in the pilots. At the moment of writing, we are looking to donate this money to NGOs conducting local development programs in these districts.

tion (Between Rounds) $_{ir}$ is a dummy that indicates that the participant received the information about the identity of the donor between the two rounds of the games. ω_{ir} are strata, survey round, and game round fixed effects, as well as participants giving in the non-political versions of the games (to control for the mechanical income effect coming from the fact that the treatment involved giving participants money) and for the Income Reporting Game a dummy for whether they were audited in a previous round. When individual fixed effects are included, γ_1 isn't estimated, as this is time-invariant.

The games' results are shown in Table 7. The first four columns show the results for the Government Fund Game, and the last four for the Income Reporting Game. For each game, results are shown for the entire sample (first two columns) and for the sample that initially believed the aid did not come from an international donor (last two columns). Odd columns don't include individual fixed effects, while even columns include individual fixed effects.

Table 7: Lab Games Outcomes (500 PKR Endowment Each)

	Government Fund Game (Effectiveness)				Income Reporting Game (Legitimacy)			
	All Sample (1)	(2)	Wrong Beliefs (3)	(4)	All Sample (5)	(6)	Wrong Beliefs (7)	(8)
γ_1 : Received Aid	-0.118 (0.917) [1.000]		-2.989** (1.460) [0.148]		0.438 (3.293) [1.000]		0.401 (5.373) [0.987]	
γ_2 : Donor Info (Between Rounds)	-1.976 (1.632) [0.643]	-2.018 (1.691) [0.019]	-3.295 (2.380) [0.467]	-2.108 (2.333) [0.110]	0.524 (4.925) [1.000]	-5.438 (5.378) [0.037]	1.075 (7.808) [0.987]	0.224 (7.775) [0.998]
γ_3 : Received Aid $\times$ Donor Info	6.435*** (1.851) [0.006]	5.642*** (1.811) [0.001]	7.990*** (2.665) [0.010]	7.277*** (2.360) [0.001]	-2.905 (5.887) [0.961]	0.687 (5.773) [0.779]	-2.934 (8.622) [0.966]	-0.243 (7.566) [0.998]
Constant	17.560*** (0.899)	28.074*** (1.226)	19.257*** (1.558)	32.698*** (2.637)	34.768*** (2.530)	57.236*** (3.104)	40.336*** (4.632)	67.273*** (5.432)
Observations	9,454	9,454	4,700	4,700	9,400	9,398	4,669	4,659
R^2	0.594	0.771	0.592	0.822	0.626	0.809	0.549	0.848
Rounds 1+2 Data	✓	✓	✓	✓	✓	✓	✓	✓
Individual Fixed Effects	×	✓	×	✓	×	✓	×	✓
$\gamma_2 + \gamma_3$	4.459***	3.623***	4.696**	5.169**	-2.381	-4.751	-1.859	-0.019

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses. Romano-Wolf multiple-testing adjusted p -values (Romano and Wolf, 2005), based on 1000 replications, are reported in square brackets, with adjustments applied jointly to Columns 1 and 5, Columns 2 and 6, Columns 3 and 7, and Columns 4 and 8. Robust standard errors in parentheses. Controlling for strata fixed effects (BISP receipt), survey round fixed effects, and game round fixed effects. For government fund game, controlling for giving to sports fund. For reported income game, controlling for income reported to NGO and whether participant has been audited in previous round.

Columns 1-4 present results for the government fund game, where each column shows the amount of the 500 PKR endowment that a participant allocates to the government fund. Columns 5-8 report results for the reported income game, where each column shows the amount of the 500 PKR income that a participant chooses to disclose to the government for a 25% tax. Within each set, the first two columns present the full sample, while the other two focus on participants who incorrectly believed the aid did not come from an international donor, with even-numbered columns including individual fixed effects.

The results for the Government Fund Game mirror those obtained using the self-reported measures: The combination of aid and information significantly and substantially increases giving to the Government Fund. This holds even when exploiting only within-individual variation. The combination leads to 5.64 PKR more given to the Gov-

ernment Fund (column 2), or $\sim 20\%$ of the control mean. These effects are concentrated among participants who wrongly believed the money did not come from an international donor (columns 3 and 4).

Similarly, the results for the Income Reporting Game mirror those when using self-reported measures of legitimacy: Neither the aid, nor the information, nor the combination of these has a significant effect on participants' income reported to the government.

Overall, the results using the incentivised measures of government effectiveness and legitimacy mimic those from the self-reported measures: The combination of the aid and the information improves participants' views of the government, particularly its effectiveness, while not affecting its legitimacy.

Political Participation & Actions: Having analysed participants' views and opinions towards the government, I now turn to three costlier types of political behaviours. The results are presented in Table 8. The first three columns look at participants' political participation within their households. In particular, I look at whether participants have tried to influence a male household member to raise or act on a community issue (column 1) or encouraged another household member to attend a community meeting (column 2). Given the mobility and political constraints women face in these communities, these are more likely ways in which they could express their political opinions. Column 3 shows an index summarising these measures. Columns 4 to 8 look at participants' participation outside the household, including speaking to a local leader about a community issue (column 4), participating in a women's group (column 5), contributing resources to a community initiative (column 6) or helping mobilise women (column 7), with a summary index in column 8. The last column looks at a real, costly political action: Participants were asked whether they agreed to sign a petition by a local NGO asking for more rights for women in their community. If they agreed, then their names would be made publicly available. This represents actual behaviour that can be captured in the survey, rather than self-reported political participation.

Overall, the results suggest that neither the aid, nor the information, nor the combination of both affects participants' political participation, either inside or outside their households. These are precisely-estimated null effects: The SEs on the indices for political participation inside and outside the household would pick up effects of around 0.065σ at the 10% level, around half the effect of what was observed for the aid plus information group in terms of the political attitudes index. Receiving the information or the combination of the aid and the information does lead to significant increases in signing the petition, but the effects are small and only marginally significant. In the PAP, I pre-specified that it was unlikely these outcomes would change, given the short-term nature of the aid and the light-touch approach to the information intervention. It is difficult to tell whether the lack of effects is due to the constraints

Table 8: Effect on Political Participation Outcomes – All Sample

	Inside HH			Outside HH				Actual Behavior	
	Influenced Male Member Raise Issue (1)	Encouraged HH Member Attend Meeting (2)	Anderson Index (3)	Spoke Local Leader Issues (4)	Participated Women's Group (5)	Contributed Resources Comm. Initiative (6)	Helped Organise Women (7)	Anderson Index (8)	Signed Petition Asking Rights (9)
β_1 : Control $\times$ Donor Info	0.013 (0.014) [0.864]	-0.003 (0.013) [0.989]	0.018 (0.044) [0.984]	0.011 (0.012) [0.955]	-0.008 (0.008) [0.951]	-0.006 (0.016) [0.999]	0.030* (0.016) [0.488]	0.017 (0.044) [0.999]	0.037* (0.020) [0.171]
β_2 : Received Aid $\times$ No Info	0.006 (0.013) [0.984]	0.010 (0.013) [0.924]	0.030 (0.046) [0.935]	0.003 (0.012) [0.999]	-0.018** (0.008) [0.200]	0.015 (0.016) [0.955]	0.013 (0.016) [0.968]	-0.016 (0.041) [0.999]	0.017 (0.020) [0.412]
β_3 : Received Aid $\times$ Donor Info	-0.001 (0.011) [0.989]	0.001 (0.011) [0.995]	-0.001 (0.037) [0.995]	-0.002 (0.010) [0.999]	-0.011 (0.007) [0.666]	0.006 (0.014) [0.999]	0.007 (0.014) [0.999]	-0.019 (0.037) [0.999]	0.031* (0.017) [0.171]
Constant	0.099*** (0.009)	0.090*** (0.008)	-0.010 (0.029)	0.073*** (0.008)	0.041*** (0.006)	0.160*** (0.011)	0.151*** (0.010)	-0.011 (0.029)	0.606*** (0.014)
Observations	4,724	4,726	4,726	4,722	4,724	4,724	4,722	4,726	4,716
R^2	0.035	0.036	0.040	0.030	0.040	0.057	0.047	0.050	0.164
$\beta_2 = \beta_3$	0.565	0.451	0.457	0.642	0.292	0.551	0.668	0.930	0.448
$\beta_1 = \beta_3$	0.270	0.730	0.628	0.228	0.738	0.417	0.132	0.354	0.745

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses. Romano-Wolf multiple-testing adjusted p -values (Romano and Wolf, 2005), based on 1000 replications, are reported in square brackets, with adjustments applied to three groups: Columns 1-3, Columns 4-8, and Column 9. Controlling for strata fixed effects and survey round fixed effects.

Columns 1-3 measure political participation inside the household. Column 1 indicates whether the participant tried to influence a male family member (husband, father, brother) to raise or act on a community issue. Column 2 indicates whether the participant encouraged or helped a household member to attend a community meeting or take action on a community issue. Column 3 is the Anderson (2008) index combining Columns 1 and 2.

Columns 4-8 measure political participation outside the household. Column 4 indicates whether the participant spoke to a local leader (e.g., neighborhood elder, religious scholar, social worker) about issues affecting the community. Column 5 indicates whether the participant participated in a women's group, community organisation, or political platform to address challenges faced by women. Column 6 indicates whether the participant contributed resources (money, food, or labor) to support a community initiative, even if not directly participating. Column 7 indicates whether the participant helped other women in the community by sharing information, organising meetings, or encouraging their participation. Column 8 is the Anderson (2008) index combining Columns 4-7. Column 9 indicates whether the participant agreed to sign a petition by a local NGO asking for more rights for women in the community.

that women in these communities face, the intervention itself, or truly a lack of effect of humanitarian aid on recipients' political participation.

Mechanisms: The results so far indicate that the combination of the aid and the information about the fact that the donor is foreign improves participants' views on the government, in particular its effectiveness. These results are confirmed using both self-reported survey measures and incentivised lab game outcomes. This can appear counterintuitive: Why would telling people that the aid came from an international donor increase support for the *national government*? In this Section, I discuss potential mechanisms that could explain this puzzling result.

One reason could be that the participants (wrongly) credit the government for the aid, and hence become more positive about the government after receiving the aid. In fact, the literature has found that development programs are prone to credit capture and misattribution (see, e.g., [Guiteras and Mobarak, 2015](#); [Cruz and Schneider, 2017](#)). However, this is unlikely to be the case for a few reasons. First, the information explicitly states that “the Pakistani government played no role in this project”, and the results in Table 3 show that the intervention succeeded in correcting participants' beliefs about the source of the funds. Second, the effects are present among both the participants who wrongly believed the aid did not come from an international donor, and those who correctly believed the aid came from an international donor. Third, if

the results were due to misattribution, we would see an effect among the participants who received the aid but not the information, which doesn't seem to be the case. Thus, credit misattribution seems unlikely to explain the results.

Another potential explanation is that participants' lives are better because of the aid (as shown in Table 2), and this makes them, in general, more positive towards the government. However, this also does not seem to be the case: If this were the case, then participants who receive the aid but not the information (whose lives improved the same as those receiving the combination) would also show a similar improvement in their views towards the government. However, this is not the case.

The mechanism seems to be something more subtle: The combination of aid and information appears to operate through a signalling mechanism about Pakistan's institutional trajectory. When recipients learn that international organisations are actively operating in Pakistan, this conveys important information beyond the immediate humanitarian assistance. In fragile states, the presence of international organisations – not just humanitarian actors who often work in the most challenging contexts – signals a level of stability and opportunity that recipients cannot easily observe themselves. International organisations undertake careful assessments of political risk, security, and operational feasibility before committing resources. Their willingness to invest and operate in Pakistan thus serves as a positive signal that the country maintains sufficient governance capacity and future prospects to merit international engagement. This presence suggests that external actors view Pakistan as a place where meaningful work can be accomplished, not merely as a site of irreversible state failure.

The aid component is crucial for this mechanism because it transforms cheap talk into costly action. While information alone might be dismissed as propaganda or empty promises, the actual delivery of resources – which significantly improved recipients' lives – demonstrates that international actors are not merely expressing optimism but are backing their assessments with concrete investments. This distinction matters particularly in contexts where citizens have extensive experience with unfulfilled promises from both domestic and international actors. The successful delivery of aid through formal channels, even without government involvement, may signal that Pakistan's institutional infrastructure – banking systems, telecommunications, and basic security – remains functional enough to support development activities.

Table 9 shows evidence in support of this mechanism, which shows that the combined treatment specifically enhances optimism about future political institutions. The first two columns look at the whole sample, while the last two include only those participants who believed the aid did not come from an international donor. The outcome in the odd columns is participants' likelihood that the political system will change in the next two years (from 0 to 10), and in the even columns, whether participants believe the political system will be better for women like them in the next two years. Recipients appear to update their beliefs about Pakistan's trajectory: Participants who

receive the aid plus the information are significantly more likely to believe that the political system will change in the next two years (column 1), and in particular to improve for women like her (column 2). The effects are also sizable: The likelihood of political change increases by 0.48 points (13% of the control mean) while the feeling that this will be better for women like them increases by 6.4 percentage points (12% of the control mean).

Table 9: Effect on Beliefs on Political Institutions

	All Sample		Believes Non. Int. Donor	
	<i>Views on the Political System in Pakistan (Next 2 Years)</i>			
	Likelihood of Change (0-10) (1)	Improve For Women Like Her (=1) (2)	Likelihood of Change (0-10) (3)	Improve For Women Like Her (=1) (4)
β_1 : Control $\times$ Donor Info	0.207 (0.145) [0.320]	0.007 (0.022) [0.758]	0.417* (0.237) [0.192]	0.045 (0.037) [0.314]
β_2 : Received Aid $\times$ No Info	0.430*** (0.144) [0.007]	0.015 (0.023) [0.740]	0.374 (0.243) [0.251]	0.039 (0.039) [0.314]
β_3 : Received Aid $\times$ Donor Info	0.476*** (0.123) [0.003]	0.064*** (0.019) [0.004]	0.531** (0.220) [0.057]	0.093*** (0.034) [0.032]
Constant	3.595*** (0.096)	0.517*** (0.015)	3.376*** (0.193)	0.492*** (0.031)
Observations	4,602	4,600	2,283	2,276
R^2	0.068	0.058	0.063	0.061
p -value $\beta_2 = \beta_3$	0.729	0.016	0.389	0.056

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses. Romano-Wolf multiple-testing adjusted p -values (Romano and Wolf, 2005), based on 1000 replications, are reported in square brackets, with adjustments applied jointly to Columns 1-2 and to Columns 3-4. Controlling for strata fixed effects and survey round fixed effects. At the bottom of the table, the p -value corresponds to a Wald test of the null hypothesis that the effect of receiving aid is the same with and without donor information. Columns 1-2 use the full sample. Columns 3-4 restrict the sample to participants who believed the aid did not come from an international donor. Columns 1 and 3 report effects on the perceived likelihood that the political system in Pakistan will change substantially in the next two years (0-10 scale). Columns 2 and 4 report effects on whether participants believe the political system will improve for women like them over the next two years.

The group receiving only the aid also sees an increase in the probability of change to the political system, but as columns 3 and 4 show, this effect is concentrated among participants who correctly identified the donor to be foreign. Moreover, the combination of the aid and the information is particularly effective in increasing these outcomes among participants who initially believed the aid did not come from an international donor, and hence for whom the information corrects these beliefs.

The intervention thus appears to shift expectations about Pakistan's institutional trajectory through the signal of sustained international engagement. Recipients interpret foreign organisations' willingness to operate in Pakistan as evidence that the country maintains viable prospects, updating their beliefs about where Pakistan is headed. This manifests in increased optimism about future political change and greater willingness to invest in government programs that might yield future returns. Yet, these same recipients show no change in state legitimacy—their sense of obligation to pay

taxes or obey authorities remains unchanged. This divergence likely reflects that effectiveness and legitimacy represent fundamentally different aspects of state assessment. While effectiveness concerns whether the state can deliver, legitimacy concerns whether it has the right to rule. The presence of international actors may credibly signal improving state capacity and future prospects, but cannot speak to questions of rightful authority that emerge from the relationship between state and citizens. The intervention essentially provides new information about Pakistan's international standing that shapes forward-looking expectations without altering core beliefs about political obligation.

Overall, the results in this Section suggest that humanitarian aid can serve as a tool for soft power, especially when coupled with information about the donor. This combination of aid and information improves recipients' views on the work of foreign organisations in Pakistan and their perceptions of the domestic government, by providing a positive signal about the country's institutional future. While more subdued, it also appears to foster greater respect towards individuals from different cultures and social groups.

6 Robustness Checks

6.1 Experimenter Demand Effects

One concern might be that of experimenter demand effects, or people answering in a certain way to "please" the surveyors. There are two different types of experimenter demand effects that may be present. First, there could be strategic answering by the control group, for example, by reporting that their situation is worse than it really is, in an attempt to qualify for the aid payments. Second, there could be strategic answering by the treatment group, for example, to make sure that they don't lose their aid payments.

Several factors reduce concerns about either sort of strategic answering. First, among the sample that is aware of the humanitarian program, everyone is told clearly at the registration point that 1) only 2/3 of participants would receive the aid payments, and 2) there was nothing they could do to change their status, to set expectations right from the beginning. Still, some participants might not believe that to be the case and could still have altered their responses. Thus, I set aside another control group that does not participate in the registration sessions and is therefore unaware of the humanitarian program's existence, as described briefly in Section 3 and in detail in Online Appendix C. By comparing the answers of this "pure control" group to those of the control group that knows about the program and may have an incentive to alter their answers in a particular way, I can assess whether experimenter demand effects are a concern among the control group.

I pre-specified this experimenter demand effects test in the PAP, and the results are presented in Table 10. It shows the results of the pre-specified outcomes for this test: the indices of three main humanitarian outcomes (which one would expect to be the most prone to suffer from experimenter demand effects), and the lab-in-the-field outcomes, both in the first round and combining game rounds. The sample includes only those individuals in the pure control group and in the control group that attended the onboarding sessions. As can be seen, there seem to be no statistically significant differences between the two groups, allaying concerns of strategic answering in the control group.

Table 10: Experimenter Demand – Pre-Specified Test

	Humanitarian Outcomes			Lab Games					
	Nutritional Diversity (1)	Food Security (2)	Mental Wellbeing (3)	Round 1		Rounds 1 + 2			
				Taxation Game (4)	Government Fund (5)	Taxation Game (6)	Taxation Game (7)	Government Fund (8)	Government Fund (9)
α_1 : Pure Control	−0.360 (0.546)	−0.008 (0.045)	−0.062 (0.045)	−4.018 (6.259)	−0.976 (1.749)	−4.867 (4.576)	−3.677 (4.154)	−1.103 (1.381)	−0.161 (1.274)
Reported Income NGO				0.710*** (0.018)		0.736*** (0.014)	0.751*** (0.012)		
Giving Sports Fund					0.818*** (0.028)			0.804*** (0.022)	0.780*** (0.028)
Audited in Previous Round				20.812** (9.619)		21.544*** (5.835)	18.751*** (4.942)	4.497** (2.087)	5.806*** (1.783)
Constant	34.457*** (0.319)	0.003 (0.027)	0.021 (0.026)	61.727*** (5.389)	21.113*** (1.665)	54.460*** (4.351)	50.015*** (3.541)	20.466*** (1.371)	20.504*** (1.494)
Observations	2,030	2,030	2,030	2,014	2,030	3,126	4,018	3,157	4,060
R^2	0.116	0.090	0.117	0.625	0.637	0.660	0.674	0.659	0.640
Round 1 Data				✓	✓	✓	✓	✓	✓
Round 2 Data				×	×	✓	✓	✓	✓
Including Info. Sample						×	✓	×	✓

Notes: Robust standard errors in parentheses. Controlling for strata fixed effects and survey round fixed effects.

While I do not conduct a test of experimenter demand effects for the treatment group, several pieces of evidence suggest that this is unlikely to be a concern here. First, those participants are already receiving their aid payments, so they have less of an incentive to manipulate their answers. Second, for the outcomes of interest in this study (values and political behaviours), it is difficult to reconcile the results with respondents trying to satisfy surveyors or researchers (e.g. the aid seems to make treated women *more* pro-government, even when they are explicitly told that the Pakistani government played no role in the aid and it came from an international donor). Third, in a related study conducted with a similar sample in Afghanistan (Callen et al., 2025), we conducted a rigorous test of experimenter demand effects among all treatment arms following De Quidt, Haushofer, and Roth (2018), and found no evidence of this.

6.2 Lab-in-the-field Games Comprehension

The lab-in-the-field games can, in principle, seem difficult for participants, especially considering the low education levels in the sample. This could raise concerns about participants' understanding of the games and their answers.

To address these concerns, the games were extensively piloted in three small-scale pilots conducted prior to the intervention, to create a set of guidelines and best practices for the games. The surveyors of the main intervention underwent four days of training, with a particular focus on the games, before the rollout of the intervention. During the surveys, participants played a practice round in which they were walked through the instructions step by step, and their questions were answered.

Table A2 shows that the understanding of the lab-in-the-field games is high. For each game, four comprehension questions were asked at the end of the practice rounds. If participants answered a question wrong, they would receive a short explanation of the correct answer. On average, participants answered 3.85 about the Government Fund game correctly, and 3.57 about the Income Reporting Game.¹¹ The lab-game results are robust to excluding participants who have a low understanding (i.e. those who get at least 3 out of the 4 questions for each game correctly), as seen in Table A3.

6.3 Balance & Attrition

The different treatment arms are largely balanced. Table A4 shows the balance among pre-specified heterogeneity (Panel A) and outcome variables (Panel B), at baseline (given that no baseline was collected for the pure control group, this group is excluded). The Table compares these outcomes for 1) the whole control group (regardless of information treatment assignment), 2) the group receiving the aid but not the information, and 3) the group receiving both the aid and the information. Means are shown in columns 2 to 5, p -values of pairwise comparisons in columns 6 to 8, and omnibus tests in column 9 and the last row. Overall, except for being the head of the household (which is slightly higher in the control group), all groups appear similar in these characteristics.

Attrition is also very low, as shown in Table A7. Of 4930 possible surveys across two rounds, 4727 were completed, a survey completion rate of 96%. Attrition was slightly higher in the second round of surveys (111 missing surveys versus 92 in the first round) due to flood threats during the last week of surveys. Most attrition comes from participants being unreachable (175, e.g., due to being out of the district or difficulties in locating them) rather than refusing to be surveyed (28, of which seven are partially completed surveys). While individuals in the cash treatment group are significantly less likely to attrit than those in the control group, this difference is not meaningful (23 more participants attrited in the control group than in the cash treatment group).

¹¹The same holds when looking at each survey round individually, as participants might get better in the second survey round due to more experience.

7 Conclusion

Foreign aid faces unprecedented skepticism as donors question whether humanitarian assistance serves any purpose beyond charity. With aid budgets shrinking while needs reach historic highs, establishing whether humanitarian aid can advance donor interests has become essential for sustaining political support. This paper provides experimental evidence that humanitarian assistance can indeed function as soft power, shaping recipients' values, attitudes, and political behaviours in ways that benefit donors – though success requires both effective aid delivery and clear attribution.

Through a randomised controlled trial with 2,450 vulnerable women in Pakistan, I demonstrate that humanitarian aid can achieve soft power objectives, especially when the aid provides tangible benefits, and recipients understand its foreign source. The cross-randomisation of aid receipt with information about its foreign origin reveals three key findings. First, the combination of aid and source information improves perceptions of foreign organisations by 10%, significantly more than either component in isolation. Second, this combination generates modest but significant increases in cultural tolerance. Third, foreign aid combined with source information increases support for Pakistan's government by 0.1σ – not through credit misattribution but because recipients interpret international investment as signalling improved national prospects. These results are robust to experimenter demand effects and multiple-hypothesis corrections, with survey response rates exceeding 96

The experimental design isolates the factors that drive soft power effects. Recipients who receive aid but believe it comes from domestic sources show minimal attitude change toward foreign actors. Those who learn about foreign involvement without receiving aid show limited response. Only the combination – effective aid with correct attribution – consistently achieves soft power objectives. This finding has immediate practical implications: The billions spent annually on humanitarian assistance could generate greater diplomatic returns simply by ensuring recipients understand the aid's source, provided the aid genuinely improves their welfare.

These findings establish proof of concept that humanitarian aid can serve strategic purposes, opening important avenues for future research. My intervention was deliberately minimal – a single verbal statement about the aid's source during surveys and short humanitarian assistance. Future work should examine what might amplify effects, for example, repeated messaging, visual branding, or community-level information campaigns. Research should also investigate whether donors can directionally shape specific outcomes rather than generating diffuse positive sentiment. Can tailored messaging promote particular values more effectively? Can attribution strategies be designed to influence political behaviours in predetermined directions? The heterogeneous effects observed – with impacts concentrated among those who initially misidentified the source – suggest targeting information toward populations with incorrect baseline beliefs might maximise impact.

The broader implications extend beyond humanitarian assistance to fundamental questions about influence in fragile states. As military interventions prove costly and traditional diplomacy faces constraints, humanitarian aid represents an under-utilised instrument for advancing national interests. The finding that foreign aid can strengthen rather than undermine domestic government support challenges conventional assumptions about foreign assistance creating dependency or weakening state legitimacy. Instead, when implemented effectively and attributed clearly, humanitarian aid can simultaneously address urgent human needs while advancing donor nations' strategic objectives – suggesting the perceived trade-off between moral imperatives and national interests may be false. As the international system faces mounting humanitarian crises amid declining resources, understanding how to maximise both the humanitarian and strategic value of aid becomes essential for maintaining political support for these life-saving programs.

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Online Appendix

A Additional Tables & Figures

Table A1: Persistence Information Provision on Beliefs & Certainty

	Round 1 Baseline Beliefs on Source					
	All Sample		Not Int. Donor		Int. Donor	
	Guessed Int. Donor (1)	Certainty (0-10) (2)	Guessed Int. Donor (3)	Certainty (0-10) (4)	Guessed Int. Donor (5)	Certainty (0-10) (6)
Received Information	0.117*** (0.023)	0.074 (0.129)	0.149*** (0.027)	0.123 (0.157)	0.028 (0.045)	-0.018 (0.224)
Constant	0.503*** (0.019)	8.798*** (0.107)	0.466*** (0.023)	8.706*** (0.130)	0.632*** (0.037)	9.099*** (0.184)
Observations	2,021	1,999	1,468	1,450	506	505
R^2	0.033	0.028	0.042	0.037	0.025	0.017

Notes: Robust standard errors in parentheses. Controlling for strata fixed effects. Odd columns show whether respondents correctly identify the identify of the funder (an international donor) at the beginning of the second round survey. Even columns show the certainty that respondents attach to their beliefs on the identify of the funder at the beginning of the second round survey. Sample restricted to second round of survey. The first two columns look at the whole sample, the following two columns look at those who incorrectly believed at the beginning of the first round survey that the aid came from someone other than an international donor, and the last two columns look at those who correctly believed at the beginning of the first round survey that the aid came from an international donor.

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Table A2: Share of Correct Questions in Games' Comprehension Questions

	Overall (1)	Control (2)	Treated (3)	<i>p</i> -value (4)
Panel A. Fund Game				
Q1. # of Funds	97.85	97.66	97.99	.551
Q2. Income Available	95.16	95.39	94.98	.488
Q3. Gov. Gets if 100 PKR Donated	94.36	93.46	95.03	.062
Q4. Sports Club Gets if 100 PKR Donated	97.28	96.4	97.94	.008
Total Correct Fund Game	3.85	3.83	3.86	.114
Panel B. Income Reporting Game				
Q1. Income Available	96.28	95.73	96.69	.153
Q2. Taxes Per 100 PKR Reported	86.45	86.25	86.6	.837
Q3. Share Black Beans (Audit)	89.43	89.12	89.66	.7
Q4. Penalty if Black Bean	85.01	83.58	86.09	.039
Total Correct Tax Compliance Game	3.57	3.55	3.59	.139

Notes: *p*-value estimated controlling for stratification fixed effects, and with robust standard errors. Those who don't answer the question are marked as having answered wrong.

Table A3: Lab Games Outcomes (500 PKR Endowment Each) – Excluding Those with Low Understanding

	Government Fund Game (Effectiveness)				Income Reporting Game (Legitimacy)			
	All Sample (1)	(2)	Wrong Beliefs (3)	(4)	All Sample (5)	(6)	Wrong Beliefs (7)	(8)
γ_1 : Received Aid	-0.398 (0.920)		-3.699** (1.471)		0.219 (3.453)		-0.064 (5.732)	
γ_2 : Donor Info (Between Rounds)	-1.919 (1.647)	-2.151 (1.711)	-3.321 (2.406)	-2.132 (2.375)	0.815 (5.211)	-4.688 (5.704)	0.277 (8.379)	1.725 (8.225)
γ_3 : Received Aid $\times$ Donor Info	6.593*** (1.860)	5.860*** (1.821)	8.450*** (2.681)	7.761*** (2.372)	-2.488 (6.242)	2.663 (6.095)	-2.248 (9.306)	0.389 (8.001)
Constant	17.894*** (0.900)	28.306*** (1.239)	19.977*** (1.560)	32.767*** (2.697)	34.385*** (2.657)	59.020*** (3.364)	41.162*** (4.937)	74.084*** (6.288)
Observations	9,308	9,308	4,594	4,594	8,546	8,541	4,139	4,127
R^2	0.593	0.773	0.591	0.820	0.624	0.821	0.545	0.858
Rounds 1+2 Data	✓	✓	✓	✓	✓	✓	✓	✓
Individual Fixed Effects	×	✓	×	✓	×	✓	×	✓
$\gamma_2 + \gamma_3$	4.674***	3.708***	5.129***	5.63***	-1.672	-2.025	-1.971	2.113

Notes: Robust standard errors in parentheses. Controlling for strata fixed effects (BISP receipt), survey round fixed effects, and game round fixed effects. For government fund game, controlling for giving to sports fund. For reported income game, controlling for income reported to NGO and whether participant has been audited in previous round.

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Table A4: Balance at Baseline

		Means				p-values			
	# Obs. (1)	All Sample (2)	Control Group (3)	Aid Only Group (4)	Aid+Info Group (5)	Control vs. Aid Only (6)	Control vs. Aid+Info (7)	Aid Only vs. Aid+Info (8)	Omnibus Test (9)
Panel A. Het. Vars									
Married	2105	0.86	0.87	0.87	0.84	0.908	0.117	0.278	0.265
Above median age	2105	0.52	0.54	0.56	0.5	0.443	0.149	0.053	0.099
No education	2105	0.84	0.82	0.85	0.84	0.055	0.372	0.298	0.188
Household's head	2100	0.51	0.55	0.52	0.49	0.259	0.009	0.272	0.033
Above med. HH size	2105	0.61	0.63	0.62	0.6	0.669	0.212	0.542	0.456
Fin. decision-maker	2105	0.2	0.2	0.2	0.21	0.842	0.485	0.721	0.794
Panel B. Outcomes									
Days skipping meals	2105	0.59	0.59	0.58	0.6	0.931	0.825	0.661	0.927
Children skipped meals	2105	0.15	0.15	0.15	0.16	0.633	0.599	0.438	0.668
Days sleep hungry	2105	0.46	0.47	0.42	0.46	0.235	0.901	0.218	0.357
Days no eating	2105	0.25	0.25	0.25	0.25	0.983	0.72	0.735	0.914
Life satisfaction (0–10)	2105	4.47	4.52	4.53	4.41	0.995	0.293	0.419	0.501
Satisfied fin. sit.	2101	0.36	0.38	0.38	0.34	0.796	0.164	0.142	0.213
Happy	2101	0.18	0.19	0.18	0.16	0.876	0.162	0.387	0.327
Improved econ. sit.	2100	0.14	0.14	0.14	0.14	0.992	0.792	0.983	0.964
Omnibus test				0.634	0.072				

Notes: p -values are from regressions restricting the sample to only the groups being compared (columns 6–8) or all groups (column 9), controlling for stratification fixed effects and clustering the errors at the individual level. Column 9 shows an omnibus test in a regression of the variable considered on the two treatment indicators, testing the equality of the coefficients of the two treatment groups. The last row shows p -values of an omnibus test of the control group vs. the aid only group (column 4) or vs. the aid with donor information group (column 5), in which the treatment dummy is regressed on all variables considered, and then a joint test of significance of all the coefficients is estimated (controlling for stratification fixed effects).

B Timeline & Pilots

Below is a summary of key activities and milestones for the project, listed in chronological order.

May–June 2024	Conducted first pilot in Multan, Punjab, with FDO as the local implementing partner, and Easypaisa as the partner mobile banking company.
July–August 2024	Carried out second pilot conducted in Rajanpur, Punjab, with AGAHE as the local implementing partner and JazzCash as the partner mobile banking company.
September–October 2024	Executed third pilot in Shikarpur, Sindh, with SRSO as the implementing partner and JazzCash as the mobile banking partner.
December 2024–January 2025	Held a two-day workshop on December 17 and 18 at the SRSO Head Office in Sukkur to train the SRSO field teams on the key components of the study, as well as the tools used for monitoring participant identification processes and data collection—such as the Google Sheets and SurveyCTO Collect mobile apps. The PWRs were launched on December 23 and concluded on January 10, except in Sukkur, where the final PWR was organized on January 18.
January 2025	Administered PSC surveys from January 13 to 28, with follow-up visits continuing until January 31 to cover missed participants in each community.
February 2025	Kicked off onboarding sessions on February 10. The last official session was on February 25. Makeup sessions were held through February 28 to accommodate participants who couldn't open their JazzCash wallet or missed their original session.
April 2025	CERP trained its survey teams from the target districts in Shikarpur from April 7 to 11. The first aid payments were disbursed on April 11 through the JazzCash disbursement portal as planned. The first round of follow-up surveys was rolled out on April 14.
June 2025	Second aid payments were made in two batches: the first on June 2 and the second on June 4. The first survey round was wrapped up on June 26.
July 2025	Following the disbursement of the third aid payments on July 3, the second round of surveys was launched on July 7, after on-line refresher sessions on the survey instrument for enumerators, held between July 2 and July 4.

August 2025

The fourth and final aid payments were transferred on August 7. The second round of follow-up surveys was concluded on September 5.

As a starting point for this project, I conducted three small pilots in three different districts: Multan, Rajanpur, and Shikarpur. In each district, we selected two communities, onboarding more or less 50 economically vulnerable women from each. Each pilot had a total sample size of no more than 100 participants. On average, approximately 94 women participated in each pilot. After identifying and registering participants from the target population, they were randomised into one of three equal groups: a control group receiving no aid, and two treatment groups receiving aid either through their national identity cards (equivalent to cash) or via mobile wallets. To analyse the effects of the intervention, a survey was conducted after the aid was delivered, focusing on humanitarian needs, mental wellbeing, (female) empowerment, and political attitudes.

The goal of the pilots was threefold:

1. to test the logistics and working dynamics with the different partners involved in the project,
2. to evaluate the feasibility of disbursing aid payments through two different channels (mobile wallets and national identity cards, called CNICs in Pakistan), and
3. to test the survey instruments and the main lab-in-the-field games that I tended to use in the full study.

These pilots provided valuable insights that significantly informed about the potential challenges, bottlenecks, and points of failure before the full project implementation. The first pilot took place in Multan, Punjab, where I discovered that 1) the local implementation partner lacked tools to monitor fieldwork and track attendance during the participant identification process; 2) Easypaisa, the partner mobile banking company, had no system for recording details of wallets opened successfully or issues encountered during failed wallet opening attempts; and 3) the money transfer platform, an entity affiliated with Easypaisa, was unable to disburse payments before the follow-up survey. I had the chance to come up with solutions to these issues, such as finding a new mobile banking company, JazzCash, and testing them in the second pilot, which proved instrumental in identifying the key problems in opening mobile wallets for participants during onboarding sessions and in refining the monitoring tools further. The main challenges participants faced in opening mobile wallets included not owning a smartphone, already owning five SIM cards (the maximum one can own), holding a dormant or suspended mobile wallet, having an outstanding loan on their existing mobile wallet, failing biometric verification due to weak fingerprints, and lacking a valid CNIC. By the third pilot, mobile wallet registration was the highest among the three pilots (possibly because the participants were informed in advance about the specific issues that would prevent them from opening wallets with JazzCash and were ad-

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vised to resolve them before onboarding), humanitarian aid was delivered before the follow-up survey, and the participant response rate was 100 percent—indicating that the solutions implemented were effective.

The pilots achieved their goal of identifying problems and informing the full study. Based on the results of the last two pilots (where aid was delivered before the surveys), it was evident that:

- the full project will be well powered to detect relevant treatment effects,
- humanitarian aid plays an important, large and immediate effect on the needs of those most vulnerable, and
- the direction of the treatment effects were in line with expectations.

C Participant Identification and Onboarding

To identify participants living in conditions resembling those of individuals in dire need of humanitarian assistance, the target population was defined as economically vulnerable women residing in urban areas. A two-step approach was used to identify eligible participants. First, households were listed and ranked into four socio-economic groups using the PWR exercises conducted in 12 communities per district. Second, the 150 lowest-ranked households in each community were selected to construct PSCs, which were then used to identify the 70 most vulnerable women. Of the 70 women, 60 were later introduced to the humanitarian aspect of the project and were part of the main sample, while the remaining 10 were not and were designated as the pure control group. These steps were carried out across three districts: Kamber, Shikarpur, and Sukkur. The project aimed to onboard 2,100 participants (700 per district) into the main RCT, with an additional 360 participants (120 per district) assigned to the pure control group.

PWR is a simple, community-based method where participating households work together to outline perceived differences in wealth and living conditions among community members. Based on these differences, they define four distinct socio-economic groups, decide which group they belong to, assign groups to absent households they know well, and endorse each other's group assignments. This process provides a broad overview of the community's living conditions based on their own views of who is doing well and who is struggling, which is important because each community may define poverty differently. While this exercise offers a useful approximation of a household's socioeconomic status, it is inherently subjective and may not be highly accurate. Nevertheless, it serves as an effective first step for narrowing the pool of households before collecting PSCs, a more rigorous but resource-heavy targeting tool.

PSC is a tool that assesses the likelihood of poverty at the household level, using proxy measures of socio-economic indicators that are easy to collect and can be objectively verified through an interview conducted at the respondent's home. The project employed its customized version by the Government of Pakistan that has been deployed in flagship programs to target potential ultra-poor households. The PSC survey collects data on indicators related to a household's living conditions, such as access to basic amenities such as sanitation, ownership of durable goods, housing type, the education level of the head of the household and household demographics. Each indicator is assigned a numeric weight, which is used to calculate the household's PSC score. The PSC complements the PWR by adding another layer of precision to the selection process.

PWRs and PSCs were carried out by the implementing partner, SRSO, across all three districts. The PWRs involving female household representatives, conducted during the project, saw an average participation of around 260 households per community, reflecting strong community involvement. Given the project's focus on extremely

poor communities, 52.19% of participants were classified in the lowest socioeconomic bin, while a further 42.23% were placed in the second lowest. Only one community in Sukkur had to be dropped due to low attendance and the PWR was instead conducted in a nearby community with higher participation. In every other location, there was no major cause for concern. Based on the PWRs, 150 households per community were selected to administer at least 130 PSC surveys.¹² Based on the poverty scores, the 70 individuals with the lowest scores in each community were considered representative of the target population, after excluding certain individuals according to a pre-specified, IRB-approved screening criteria outlined in Table A5. Women in the PSC who had no access to a mobile phone (11), lacked a valid CNIC or had an expired one (54), were under 18 or over 65 years of age (284), or had difficulties with hearing, listening, seeing, remembering, concentrating, or moving without assistance (58) were excluded from the pool used to draw the final sample.¹³ This screening was crucial for selecting participants who were ready to take up the intervention, primarily those with a valid CNIC and access to a mobile phone.

Table A5: Screening Criteria

Reason	# Individuals
No CNIC	7
CNIC expired	47
Outside age range	284
No access to phone	11
Disability	58
<i>Total excluded</i>	397
Final sample that meets criteria	4624

Notes: An individual can satisfy several of these restrictions at the same time.

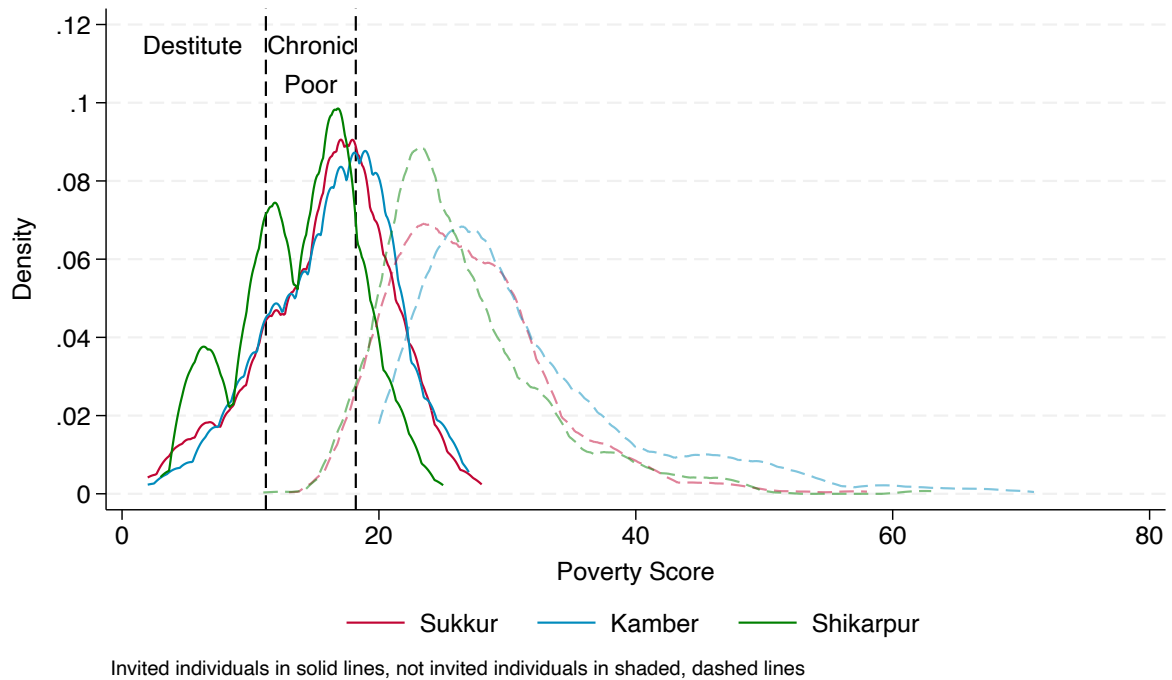
Figure A1 shows the distribution of the poverty scores across the three districts, for those who were invited to proceed to the next stage (solid lines) and those who were not (dashed lines). As can be seen, participants seem indeed very poor, with the mean and median poverty score among invited participants below the line considered “chronically poor” (a score of 18) in each of the three districts, and more than a fifth considered “destitute”. Encouragingly, the distributions in the three districts look similar. It also appears that the PWR is effectively capturing the socioeconomic differences as intended: households in the lowest socioeconomic bin have the lowest average

¹²When there were more than 150 households in the lowest bin, 150 were randomly selected from within that group. If there were fewer than 150 households in the lowest bin, then all households in that bin were included, along with a random sample from the second-lowest bin to reach the target of 150.

¹³Note that some participants meet multiple conditions simultaneously, which is why the sum does not add up to the total number excluded.

poverty score (20.75), followed by those in the second-lowest bin (22.83) and those in the next higher bin (26.63).

Figure A1: Distribution of Poverty Scores by Onboarding Status



To recruit participants for the field experiment, women identified through the PSC were invited to a neutral location in each community to complete a set of activities required for participation in the project, while ten women in each community were randomly set aside as part of the pure control group and were not invited. Overall, 360 women were assigned to the pure control group (120 per district) and 2160 were invited to participate in onboarding sessions (720 per district), in groups of 30. Two onboarding sessions took place per community, both on the same day (morning and afternoon). In these sessions, four activities took place. First, participants learned about the project (what it is about, who will receive the humanitarian aid and how, and the source of funding). This is the first time they hear that the project is about humanitarian aid, and that two-third of them will receive aid payments—with monthly payments of 25 USD each. Second, participants provided their consent to participate in the project, following a consent form approved by LSE’s and CERP’s IRB committees. Third, participants completed a short baseline survey, primarily focused on demographic characteristics. Lastly, participants opened their mobile wallets with the help of a team from JazzCash and were guided on how to use these wallets while being handed a brochure.

Table A6 shows that participants selected for the experiment are much poorer and more vulnerable than those who were not selected, based on the PSC data (columns 1-4), as intended. Column 4 reports the p -values from the tests of differences between individuals not selected (column 2) and those selected (column 3), which includes both

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individuals invited to the onboarding sessions and those assigned to the pure control group. The results indicate that participants chosen for the experiment generally have less favourable socioeconomic characteristics than those not selected, suggesting that the selection process effectively targeted the poorest and most vulnerable.

The remaining columns focus on the balance between individuals invited to the onboarding sessions (column 6) and those not invited to the onboarding sessions (i.e., the pure control group; column 5), with p -values for differences between the two groups reported in column 7. Only one of the differences is significant at the 10% level, in line with what would be expected by random chance considering 11 outcomes analysed, suggesting that there is balance between the pure control group and those invited to the onboarding sessions at the time of onboarding assignment.

Table A6: Poverty Score Card Descriptives

	Sample Selection				Onboarding Assignment		
	All Sample (1)	Not Selected (2)	Selected Sample (3)	p -value (2 vs. 3) (4)	Pure Control (5)	Invited to Onboarding (6)	p -value (5 vs. 6) (7)
Age	40.73 (11.77)	41.26 (12.68)	40.29 (10.94)	.006	40.44 (10.89)	40.26 (10.95)	.774
Has no education	0.76 (0.43)	0.66 (0.47)	0.84 (0.37)	0	0.84 (0.37)	0.84 (0.37)	.965
Is head of household	0.51 (0.5)	0.49 (0.5)	0.52 (0.5)	.105	0.56 (0.5)	0.51 (0.5)	.103
Children HH members	3.02 (2.2)	1.92 (1.72)	3.93 (2.14)	0	4.04 (2.34)	3.91 (2.1)	.314
Elderly HH members	0.36 (0.6)	0.31 (0.55)	0.39 (0.63)	0	0.39 (0.61)	0.39 (0.63)	.989
BISP beneficiary	0.38 (0.49)	0.3 (0.46)	0.46 (0.5)	0	0.48 (0.5)	0.45 (0.5)	.321
Owns feature phone	0.56 (0.5)	0.53 (0.5)	0.58 (0.49)	0	0.62 (0.49)	0.58 (0.49)	.156
Owns smartphone	0.18 (0.38)	0.22 (0.41)	0.14 (0.35)	0	0.13 (0.34)	0.14 (0.35)	.502
Has mobile money account	0.08 (0.27)	0.09 (0.29)	0.07 (0.26)	.004	0.05 (0.22)	0.07 (0.26)	.06
Poverty score	21.27 (9)	28.27 (7.7)	15.42 (4.91)	0	15.05 (5.37)	15.48 (4.82)	.147
Ranked bottom bin in PWR	0.76 (0.43)	0.76 (0.42)	0.76 (0.43)	.901	0.76 (0.43)	0.76 (0.43)	.834

Notes: Standard deviations are reported in parentheses. Column 1 shows statistics for the full sample. Column 2 includes individuals not selected for the experiment. Column 3 includes individuals selected for the experiment, combining both those invited and those not invited to onboarding. Column 4 reports p -values testing differences between individuals selected for the experiment and individuals not selected. Column 5 restricts to the pure control group (individuals selected for the experiment but not invited to onboarding). Column 6 includes individuals invited to onboarding. Column 7 reports p -values testing differences between individuals in the pure control group and those invited to onboarding.

Out of the 2160 participants invited to the onboarding sessions, 2105 agreed to participate and met the eligibility criteria (exclusions at this stage are discussed below).

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These 2,105 were randomly assigned to three groups: a control group (not receiving humanitarian aid), a cash group (receiving aid via their CNIC), and a digital group (receiving aid through their mobile wallets). These participants, along with the 360 women in the pure control group, comprise the final sample for this project.

D Attrition

For this study, attrition refers to the exclusion of participants at various stages of data collection. That is, individuals who progressed to a subsequent stage but did not continue participating for any reason were considered attrited. Tracking attrition is essential for assessing the internal validity of the study, as systematic differences between those who remain in the sample and those who drop out can bias estimates of treatment effects. Table A7 summarizes attrition patterns across treatment arms and survey rounds.

Attrition in this study occurred at three main stages, and at each stage, it occurred along two primary margins, which are labeled as “Not Surveyed” and “Refused”. The former refers to those who missed the survey for reasons other than outright refusal, while the latter refers to explicit refusals.:

1. During the onboarding sessions that included the baseline survey, when some individuals were unreachable, did not show up, or failed to meet the eligibility criteria.
2. At the first follow-up survey, when a portion of the baseline sample could not be reached or were unavailable for reasons other than refusal, while others explicitly declined to continue participating.
3. In the second follow-up survey, when additional loss occurred among those who had previously participated in the first follow-up, as some participants moved out of town to a safer location due to the risk of flooding from the Indus River.

At the onboarding stage, 55 out of 2,160 identified participants were excluded from the project. Of these, 18 did not attend the onboarding sessions, 6 women attended but refused to participate, 18 were found to have expired CNICs at the time of onboarding, and 13 did not meet the original eligibility criteria, such as being above the age cutoff or having severe impairments. There were some differences observed between the 55 women who were excluded and the 2105 who make the final sample. Having said that, there is no consistent pattern found. The excluded group appears better off on some indicators (e.g., they report fewer instances of children skipping meals and are less likely to be BISP beneficiaries), but worse off on others (e.g., they are more likely to be categorized as deprived). Given this, and the small sample that was excluded, this is not considered a cause of concern.

The first follow-up survey was launched about one and a half months after the onboarding phase and lasted about two and a half months. Of these, 2,373 were successfully surveyed, but 92 participants (around 3.7%) were lost to follow-up. Attrition happened along two primary margins: some participants were unreachable or missed the survey owing to reasons such as travel or being unreachable, while others explicitly refused to participate. As shown in Table ??, participants in the digital group were 2.2 percentage points more likely to miss the survey (Column 1) compared to the con-

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trol group, a statistically significant difference at the 5% level. However, they were also 0.9 percentage points less likely to refuse taking the survey, significant at the 5% level. For the cash group, no significant differences in either dimension of attrition were observed. When looking at total attrition (whether not surveyed or refused), neither treatment group shows a statistically significant difference from the control group. Overall, attrition rate was low and broadly similar across treatment arms, reducing concerns about bias.

Survey participation remained high across both survey rounds, though attrition increased slightly in the second round, where 111 participants out of 2,465 (around 4.5%) did not complete the survey (37 in the control group, 26 in the pure control group, 20 in the cash group, and 28 in the digital group). The rise in attrition was primarily because many participants relocated to safer areas in response to a flood warning issued during the final week of the surveys. Importantly, attrition appears broadly balanced across treatment groups, with only small and statistically insignificant differences relative to the control group for most groups. Participants who received aid in cash were somewhat less likely to refuse or attrit than those in the control group, but the difference is minor and not a cause for concern.

Taken together, attrition rates were low in each follow-up round ($\sim 3.7\%$ in the first round and $\sim 4.5\%$ in the second round) as well as in the combined sample ($\sim 4.1\%$), with only modest differences by treatment status.

Table A7: Survey Attrition

	Round 1			Round 2			Combined		
	Not Surveyed	Refused	Not Surveyed or Refused	Not Surveyed	Refused	Not Surveyed or Refused	Not Surveyed	Refused	Not Surveyed or Refused
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
β_1 : Pure Control	0.001 (0.011)	-0.007 (0.005)	-0.006 (0.012)	0.021 (0.015)	-0.002 (0.006)	0.020 (0.016)	0.011 (0.009)	-0.004 (0.004)	0.007 (0.010)
β_2 : Aid Given Digitally	0.022** (0.010)	-0.009** (0.004)	0.013 (0.011)	-0.010 (0.010)	-0.003 (0.005)	-0.013 (0.011)	0.006 (0.007)	-0.006* (0.003)	0.000 (0.008)
β_3 : Aid Given in Cash	-0.003 (0.008)	-0.006 (0.004)	-0.008 (0.010)	-0.015 (0.010)	-0.008** (0.004)	-0.024** (0.011)	-0.009 (0.006)	-0.007** (0.003)	-0.016** (0.007)
Constant	0.027*** (0.006)	0.010*** (0.004)	0.037*** (0.007)	0.043*** (0.008)	0.010*** (0.004)	0.052*** (0.008)	0.035*** (0.005)	0.010*** (0.003)	0.045*** (0.006)
Observations	2,465	2,465	2,465	2,465	2,465	2,465	4,930	4,930	4,930
R^2	0.003	0.002	0.002	0.004	0.002	0.005	0.001	0.001	0.002
Attrited	80	12	92	95	16	111	175	28	203

Notes: Coefficients represent differences in attrition rates between each treatment arm and the control group. Robust standard errors in parentheses. refers to participants who missed the survey for reasons other than outright refusal, while refers to those who explicitly declined to complete it. Partially completed surveys (4 in Round 1 and 3 in Round 2) are coded as .

E Lab-in-the-Field Games

Two lab-in-the-field games were conducted during each follow-up survey to measure attitudes toward the state, particularly beliefs about government effectiveness and legitimacy.

Game A: Giving to a Government Fund This game presents participants with a simple but revealing decision. Each individual receives a 500 PKR endowment and is asked to decide how much of this amount they would like to keep for themselves and how much they would like to contribute to a government fund that will be used for local development projects. To incentivise contributions, each PKR given to the government fund is doubled.

The primary outcome is the amount each participant allocates to the government fund, serving as a behavioural proxy for their trust in the government and their perception of its effectiveness in delivering services or completing projects. If participants perceive that the government has the capacity and will genuinely conduct the local development project, they are more likely to contribute. Conversely, low contributions may indicate scepticism about the government's ability to utilise the fund effectively or concerns about misappropriation of resources entrusted to it. In this way, the game offers a behavioural measure of political outcomes that goes beyond self-reported attitudes.

To control for income effects, participants are later presented with another 500 PKR endowment. This time, however, they are asked to allocate it between themselves and a local sports club, a non-political entity serving as a neutral alternative to the government fund. This is the same approach as that followed in [Acemoglu et al. \(2020\)](#), also in Pakistan, so it has already been tailored to the local setting. The extent to which participants' behaviour is driven by their beliefs about the government can be isolated by comparing how much more (or less) money they contribute to the government relative to the sports club.

Game B: Reporting Income to Government. This game is implemented to measure participants' willingness to comply with government laws and mandates, which can be interpreted as an indicator of political legitimacy, following [Levi, Sacks, and Tyler \(2009\)](#). The game involves a simple taxation scenario in which participants receive an endowment (referred to as income) and decide how much of it to report to the government, which will deduct a proportion of it as tax. The game is adapted from [Blair, Marty, and Roessler \(2022\)](#), who used a similar design in Sierra Leone.

Each participant is given an income of 500 PKR and asked to decide how much of this amount to report to the government and how much to withhold. Any reported income is taxed at a rate of 25%. If a participant withholds an amount X , there is a 10% probability that they will be audited by the government, in which case they must pay a

50% penalty on the undeclared income X . With a 90% probability, they are not audited and thus retain all of the unreported income. This trade-off incentivises participants to weigh the potential gains from evading taxes against the risk of incurring a penalty.

Note that the values used in the game (a 25% tax rate and a 10% audit probability) do not reflect actual government rates or the true likelihood of being audited, which are difficult to determine. These values have been arbitrarily chosen to simplify the decision-making process and ensure that participants with limited numeracy skills can understand the incentives.

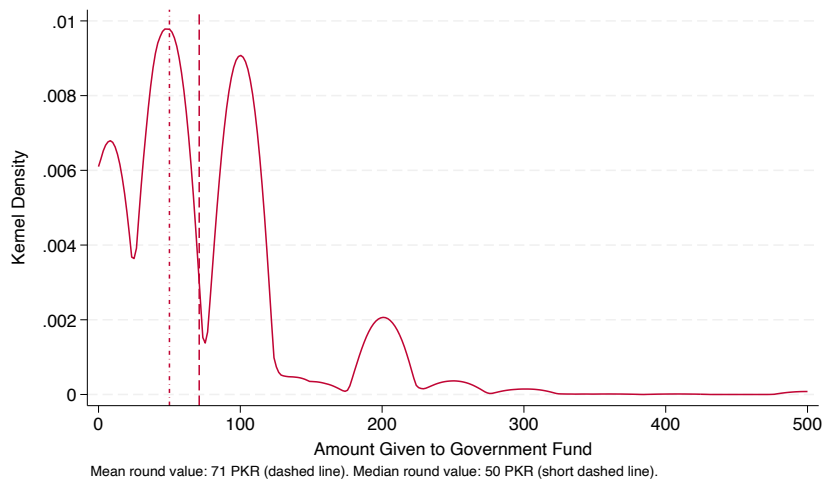
The decision to comply with or evade taxation allows us to measure legitimacy in the sense defined by [Levi, Sacks, and Tyler \(2009\)](#), who view legitimacy as the willingness to obey authorities. In this context, higher levels of reported income can be interpreted as a stronger willingness to comply with government rules, which, in turn, can be seen as an indicator of higher perceived political legitimacy. It also captures a key component of the social contract: The act of paying taxes ?. This means citizens are more likely to pay taxes if they believe the state uses these revenues to provide public goods and services that benefit everyone, reflecting a reciprocal relationship.

To control for income effects, participants also play a variant of this game, in which they face the same decision, but the context is repaying a microloan from an NGO, a non-political entity. The same tax, audit, and penalty structure applies: Repayments are expected, and undisclosed profit may be audited with a 10% probability and penalised at 50%. By comparing behaviour across the two versions of the game (government vs. NGO), the extent to which the perceived legitimacy of state authority drives compliance with government regulation can be disentangled from other influences.

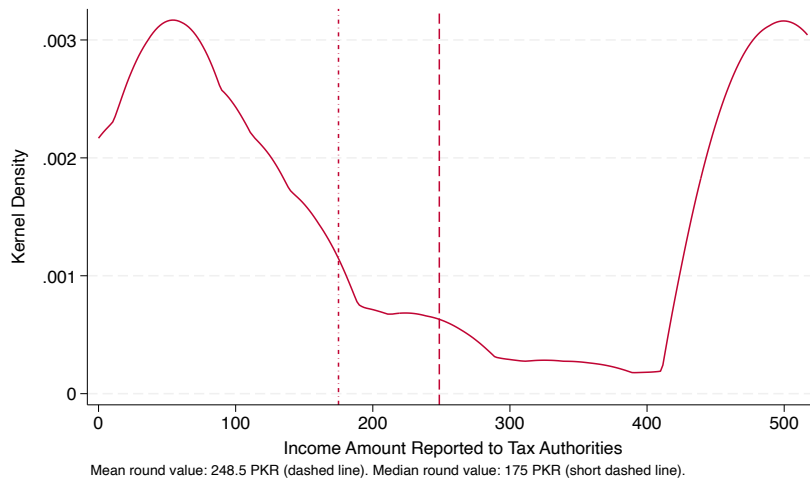
Lab Game Implementation The games are played thrice. The first time, after eliciting beliefs on the source of the funds at the beginning of the survey and before providing information on the identity of the donor. This is a practice round, in which enumerators walk participants through the games in detail and answer any questions they might have. Comprehension questions are asked at the end of the practice rounds. To ensure that participants do not feel pressured to respond in any way, enumerators hand over their tablet, which displays a slider that allows participants to decide their allocation in private. The games are then played a second time, this time for real. Finally, the games are played a third time, immediately after providing information about the source of the donor. Two allocations are randomly provided (out of the eight possible, as there are two games, a political and a non-political version of each, and two rounds), and transferred to the participants a few days after the completion of the survey, together with the 500 PKR appreciation token for completing the survey. On average, participants receive approximately 1350 PKR based on their responses.

Figure A2 shows the distribution of how much participants give to the Government Fund (Panel A) and how much they report to the government in the Reported Income game (Panel B) in the first round of the games in the first round of surveys. Both outcomes show considerable variation. On average, participants allocate 71 PKR (out of 500 PKR) to the Government Fund, with large concentrations around 0, 50, and 100 PKR, and to a lesser extent, 200 PKR. For the Income Reporting game, participants report, on average, almost half of their endowment (248.5 PKR), with a large share of participants reporting all their income.

Figure A2: Lab Game Outcomes in Round 1



(a) Government Fund Game



(b) Income Reporting Game

Notes: The Panels show participants' giving to the Government Fund (A) and their reported income to the government (B) in the first game played for real during the first round of surveys.

Below is the exact script that enumerators followed during the surveys.

Enumerator Script

Games Introduction Now that we are finished with those questions, I want to do some activities with you. In total, we will do three different activities. I will explain each activity to you, give you several examples, and answer any questions that you might have. All your answers will be kept private, as I will ask you to enter your responses in my smartphone/tablet without me seeing. All the activities will give you a hypothetical scenario, and ask you how you would allocate fictitious money in each of the scenarios. At the end of the day, one of activities we will do will be randomly picked and the allocation that you decided will be implemented. This means that while the money we are playing with is fictitious, you will get real money in a few days' time based on your responses, so make sure you answer truthfully.

Game A Instructions We are establishing two funds in your area. There is one fund that will be given to government-affiliated institutions to be used for local programs in your community. Let's call this the government fund. The second fund will be used to help the local sports club in your community. Let's call this the sports club fund. We are taking your valuable opinions because even though we are the ones making these funds, we do not yet know what the average woman in communities like this needs and how much money should be allocated to each fund. In order to ascertain your opinion, we will conduct the fund exercise. Do you have any questions?

Now I will show you how we will conduct this exercise. Each time I will provide you with 500 PKR, and you will think of it as your own money. You can decide how much money to allocate to the government fund and how much to keep yourself, based on how effective and helpful you think the government fund will be in helping people in your community. Every PKR you donate to the government, so the state will receive twice the amount of money you allocated.

To give you an example, if you decide to keep 300 PKR and donate 200 PKR to the government fund, you will receive 300 PKR, and the government will receive 400 PKR in total (two times 200). No one will know how much you decided to keep and how much you decided to give to the government fund. When we play the game, I will give you my smartphone/tablet, and then you can enter how much you want to keep for yourself, and how much to give to the government fund, without me seeing what you entered.

We will do the activity several times, each time we will do it once with the government fund, and once with the local sports club fund. Everything will be the same when we do it for the government or the local sports club fund. Do you have any questions?

Now I will ask you some questions to check whether you understood the game:

- How many funds are there, and what are they related to?
Answer: Explain that there are two, the one associated with the government, and the one associated with the local sports club.

- How much money do you have to allocate between yourself and the government or the sports club?

Answer: 500 PKR

- If you decide to allocate 50 PKR to the government fund, how much would they get?

Answer: 100 PKR, as any money allocated to the government fund will be doubled.

Game A Practice Round Now let's play a practice round, so that any additional questions can be clarified.

We are considering the possibility of establishing two funds in your area. There is one fund that will be given to government-affiliated institutions to be used to help with local programs in your community. Let's call this the government fund. The second fund will be used to help the local sports club in your community. Let's call this the sports fund. We are taking your valuable opinions because even though we are the ones thinking of making these funds, we do not yet know what the average woman in communities like this needs and how much money should be allocated to each fund. In order to ascertain your opinion, we will conduct the fund exercise.

You have 500 PKR, and can decide how much to allocate to yourself and how much to allocate to the government fund. Any money allocated to the government fund will be doubled.

Question. How much do you want to keep for yourself, and how much do you want to give to the government fund? In order for your answers to be private, I will give you my smartphone/tablet, so that you can enter the amounts you want without me seeing. Remember that this round is only practice, so let me show you how you can do this.

Question. How much to the government fund?

Enumerator instructions: Make sure you explain to the participant how the slider works.

Question. Based on your answer, this is what you and the government would get

- Government Fund: *[slider value]* PKR
- You: *[500 – slider value]* PKR

Is this what you want?

Now, you have 500 PKR, and can decide how much to allocate to yourself and how much to allocate to the sports fund. Any money allocated to the sports fund will be doubled.

Question. How much do you want to give to the sports fund?

Enumerator instruction: Hand over the tablet so that they can pick their answer

Question. Based on your answer, this is what you and the sports fund would get

- Sports Fund: [slider value]PKR
- You: [500 – slider value] PKR

Is this what you want?

Do you have any questions? This is the last time we will play this game as a test, so now is a good moment to ask any questions you might have.

Game A Comprehension Test I now want to ask you some questions about the activity we just discussed.

Enumerator instruction: Do not provide a hint to the correct answer. Select the answer that the participant has chosen without giving them any help. If participant has made a mistake, explain the right answer after you have recorded her answer.

Question. How many funds are there in this game?

Enumerator prompt if participant answers incorrectly: The answer is not [participant's answer]. Remember that there are two funds. One fund will go to the government, and the other fund will go to the local sports club. We will do this activity with each of these funds separately. Is this clear?

Question. For this activity, what is the income that you have available to divide between you and the government or sports club fund?

Enumerator prompt if participant answers incorrectly: The answer is not [participant's answer]. Remember that every time we do this activity, you will have 500 PKR. You can then decide how much of this to keep for yourself, and how much to give to either the government or the sports club. Is this clear?

Question. For every 100 PKR that you give to the government fund in this activity, how much money does the government receive?

Enumerator prompt if participant answers incorrectly: The answer is not [participant's answer]. Remember that for every PKR that you give to the government fund, the government receives twice as much money. So if you decide to allocate 100 PKR to the government fund, the government will receive 200 PKR. Is this clear?

Question. For every 100 PKR that you give to the sports club fund in this activity, how much money does the sports club receive?

Enumerator prompt if participant answers incorrectly: The answer is not [participant's answer]. Remember that for every PKR that you give to the sports club fund, the sports club receives twice as much money. So if you decide to allocate 100 PKR to the sports club fund, the sports club will receive 200 PKR. Is this clear?

Game B Instructions This was the first set of activities. We will do the real rounds of this activity later. Now, I am going to explain you the second activity, and we will do a round of practice as well. Once we have completed the practice of the second activity, we will do the real versions of each of these two activities.

Just like in real life, you must report the income you earn to the government. In this activity, you will decide how much income you earn in this activity to report to the government. Again, while this is a hypothetical scenario and your answers will be private, we will randomly pick one of the activities and implement the established allocation.

Also just like in real life, you must pay taxes on the income you report. These taxes are not pretend. At the end of the activity we will give all the taxes we collect to the government. So even though the activity is fun, it is not really a game. The tax rate for this activity is 25%. That means that for every 100 PKR you report, the government will take 25 PKR and you will keep 75 PKR for yourself.

Also just like in real life, the government will not know how much income you earn. That means you must decide whether to report all of your income, some of your income, or none of your income. The more income you report, the more the government will take from you in taxes. Do you have any questions?

Like I said before, the government will not know how much income you earn. But just like in real life, the government can decide to investigate you to find out. If the government decides to investigate you, it will compare the income you earned to the income you reported. If the government finds out that you earned more than you reported, it will punish you with a fine of 50% of what you don't report. It will also tax you on all the income you reported, at the usual 25% rate. Also just like in real life, the government will not investigate everybody. After every round of the activity you will pick a bean from this bag. That bean will tell you if you will be investigated or not. If you pick a black bean, that means you will be investigated. If you pick a white bean, that means you will not be investigated. There are 18 white beans in the bag, but only 2 black beans. That means the chance you will be investigated is 10%, or 1 in 10. Do you have any questions?

Let's do one example. Let's say that a participant called Fatima earns 500 PKR. Fatima then decides how much to report to the government, and how much not to report to the government.

Let's see what Fatima could do:

- Let's say Fatima reports all 500 PKR. In this case, the government will take 125 PKR, and Fatima will keep 500 PKR minus the 125 PKR she paid in taxes, so she gets 375 PKR.
- Let's say Fatima reports 100 PKR. There are two scenarios:

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- 9 out of 10 times, Fatima will not be audited by the government (she draws a white bean), so she will pay 25 PKR to the government and keep the remaining 475 PKR for herself.
- 1 out of 10 times, Fatima will be audited by the government (she draws a black bean). This means that she will have to pay taxes on her reported income, so 25 PKR in taxes, as well as pay a fine of 200 PKR, half of the income she did not report. This means that she will only get 275 PKR.
- Let's say Fatima reports 400 PKR. There are two scenarios:
 - 9 out of 10 times, Fatima will not be audited by the government (she draws a white bean), so she will pay 100 PKR to the government in taxes and keep the remaining 400 PKR for herself, 300 PKR that is left after paying taxes from the money she reported, and 100 PKR from the money she did not report.
 - 1 out of 10 times, Fatima will be audited by the government (she draws a black bean). This means that she will have to pay taxes on her reported income, 100 PKR in taxes, as well as pay a fine of 50 PKR, half of what she did not report. This means that she will only get 350 PKR.

Enumerator instruction: Take out the table with the different values and explain it to the participant.

You can see in the following table, for different values of what you report and what you don't report, how much money you would make if he draw a black or a white bean. Do you have any questions?

Now we will do one round of practice of the activity to make sure everything is clear. This is just practice. You will not earn any real money in this round. You can ask questions at any time during this example. If anything is unclear, please just ask and I will explain.

You will have 500 PKR, and you can report any amount you want. You cannot report more income than you actually earned. For example, you can report 0 PKR, 5 PKR, 10 PKR, all the way up to 460 PKR, 480 PKR, 500 PKR. You can report all of it, you can report none of it, or you can report any amount between all and nothing. That one is your secret.

I will give you my smartphone/tablet so that you can privately enter much you decide to report. To see who will be investigated, you will pick a bean from this bag. If you pick a black bean, that means you will be investigated. Do you have any questions?

Game B Practice Round Let's do the practice round. You earned 500 PKR, how much do you decide to report to the government? As this is a practice round, let me help you enter the value.

Question. How much do you decide to report to the government?

Enumerator instruction: Hand over the smartphone/tablet to the participant, and show them how they can enter their allocation.

Question. Based on your answer, this is what you and the government would get if you draw a white bean

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

and this is what you and the government would get if you draw a black bean

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

Is this what you want?

Question. Ok, now that you have decided how much to allocate, let's draw one of the beans to see if the government would audit you in this fictitious scenario. What is the color of the bean? *Enumerator instruction: Have the participant draw a bean.*

Enumerator prompt if participant draws a white bean: Ok, you drew a white bean. This means that the government would not audit you, so you would get the following allocation

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

If you had drawn a black bean, the government would have audited you, and you would have received

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

Enumerator prompt if participant draws a black bean: Ok, you drew a black bean. This means that the government would audit you, so you would get

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

If you had drawn a white bean, the government would not have audited you, and you would have received

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

Do you have any questions?

Game B Comprehension Test I now want to ask you some questions about the activity we just discussed.

Question. For this activity, what is the income that you have available?

Enumerator prompt if participant answers incorrectly: The answer is not [participant's answer]. Remember that every time we do this activity, you will have 500 PKR. You can then decide how much of this income to report to the government, or how much not to report to the government. Is that clear?

Question. For every 100 PKR that you report to the government in this activity, how much money does the government take?

Enumerator prompt if participant answers incorrectly: The answer is not [participant's answer]. Remember that for every 100 PKR that you report to the government, you have to pay 25 PKR in taxes. Is that clear?

Question. Out of 20 beans in the bag, how many are white?

Enumerator prompt if participant answers incorrectly: The answer is not [participant's answer]. Remember that there are 2 black beans, and 18 white beans. When you draw a black bean, you are audited and you have to pay a penalty if you did not report some income. This only happens in 1 out of 10 cases. In the remaining 9 cases when you draw a white bean, the government doesn't audit you and you can keep all the money you did not report. Is that clear?

Question. Imagine that you don't report 200 PKR to the government. If you draw a black bean, how much of those 200 PKR will the government take from you?

Enumerator prompt if participant answers incorrectly: The answer is not [participant's answer]. Remember that if for every 100 PKR that you don't report to the government, you have to pay half of it as a penalty if the government audits you. If you don't report 200 PKR, this means that you would need to pay half, meaning 100 PKR, in penalty to the government. Is that clear?

Game A Real Round One Now we will begin the real rounds. Remember, while we are talking about hypothetical money in the activities, so you should treat this money like it is real. We will play each of these activities twice, and for each activity, we will randomly pick one of your answers, and you will be transferred the money over JazzCash in the coming days.

You have 500 PKR, and can decide how much to allocate to yourself and how much to allocate to the government fund. The money allocated to the government fund will be given to government-affiliated institutions to be used to help with local programs in your community. Any money allocated to the government fund will be doubled.

Question. How much do you want to give to the government fund?

Enumerator instruction: Hand over the tablet so that they can pick their answer

You have 500 PKR, and can decide how much to allocate to yourself and how much to allocate to the government fund. The money allocated to the government fund will be given to government-affiliated institutions to be used to help with local programs in your community. Any money allocated to the government fund will be doubled.

Question. Based on your answer, this is what you and the government would get

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

Is this what you want?

Now, you have 500 PKR, and can decide how much to allocate to yourself and how much to allocate to the sports fund. Any money allocated to the sports fund will be doubled.

Question. How much do you want to give to the sports fund?

Enumerator instruction: Hand over the tablet so that they can pick their answer

Question. Based on what your answer, this is what you and the sports fund would get

- Sports Fund: [slider value] PKR
- You: [500 – slider value] PKR

Is this what you want?

Game B Real Round One You have earned 500 PKR. Just like in real life, you must report the income you earn to the government. Also just like in real life, you must pay taxes on the income you report. These taxes are not pretend. At the end of the activity we will give all the taxes we collect to the government. The tax rate for this activity is 25%. That means that for every 100 PKR you report, the government will take 25 PKR and you will keep 75 PKR for yourself. Also just like in real life, the government will not know how much income you earn. That means you must decide whether to report all of your income, some of your income, or none of your income. The more income you report, the more the government will take from you in taxes. Just like in real life, the government can decide to investigate you to find out if you did not report some of your income. If the government decides to investigate you, it will compare the income you earned to the income you reported. If the government finds out that you earned more than you reported, it will punish you with a fine of 50% of the income you did not report. It will also tax you on the income you reported. Also just like in real life, the government will not investigate everybody. After every round of the activity you will pick a bean from this bag. That bean will tell you if you will be investigated or not. If you pick a black bean, that means you will be investigated. If you pick a white bean,

that means you will not be investigated. There are 18 white beans in the bag, but only 2 black beans. That means the chance you will be investigated is 10%, or 1 in 10.

Question. How much do you decide to report to the government?

Enumerator instruction: Hand over the smartphone/tablet to the participant, and show them how they can enter their allocation.

Question. Based on your answer, this is what you and the government would get if you draw a white bean

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

and this is what you and the government would get if you draw a black bean

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

Is this what you want?

Ok, now that you have decided how much to allocate, let's draw one of the beans to see if the government would audit you in this fictitious scenario. What is the color of the bean? *Enumerator instruction: Have the participant draw a bean.*

Enumerator prompt if participant draws a white bean: Ok, you drew a white bean. This means that the government would not audit you, so you would get the following allocation

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

Enumerator prompt if participant draws a black bean: Ok, you drew a black bean. This means that the government would audit you, so you would get

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

Game B Variant Real Round One Now we will do a similar activity to the one we did before.

Imagine you received funding from a local NGO to support a small business run by you. The funding works the following way: Only those people who made a profit need to pay a share to the NGO for the profits they made. This way, those who did better pay more, just like in real life. In this activity, you will decide how much profit to report to the NGO. Again, while this is a hypothetical scenario and your answers will be private, we will randomly pick one of the activities and distribute your rewards based on its results.

It turns out you made 500 PKR in profits. You now need to decide how much of these profits to report to the NGO. Remember, even though the activity is fun, it is not really a game. Similarly to the previous activity, the NGO takes a 25% share of the profits you made. That means that for every 100 PKR you report, the NGO will take 25 PKR and you will keep 75 PKR for yourself.

Also just like in real life, the NGO will not know how much profit you made. That means you must decide whether to report all of your profit, some of your profit, or none of your profit. The more profit you report, the more the NGO will take from you as its share.

Like I said before, the NGO will not know how much profit you made. But just like in real life, the NGO can decide to investigate you to find out. If the NGO decides to investigate you, it will compare the profit you reported to the profit you made. If the NGO finds out that you made more than you reported, it will punish you with a fine of 50% of what you don't report. It will also collect its share on all the profit you reported, at the usual 25% rate. Also just like in real life, the NGO will not investigate everybody. After every round of the activity you will pick a bean from this bag. That bean will tell you if you will be investigated or not. If you pick a black bean, that means you will be investigated. If you pick a white bean, that means you will not be investigated. There are 18 white beans in the bag, but only 2 black beans. That means the chance you will be investigated is 10%, or 1 in 10.

Question. How much do you decide to report to the NGO?

Enumerator instruction: Hand over the smartphone/tablet to the participant, and show them how they can enter their allocation.

Question. Based on your answer, this is what you and the NGO would get if you draw a white bean

- NGO: [slider value] PKR
- You: [500 – slider value] PKR

and this is what you and the NGO would get if you draw a black bean

- NGO: [slider value] PKR
- You: [500 – slider value] PKR

Is this what you want?

Ok, now that you have decided how much to allocate, let's draw one of the beans to see if the NGO would audit you in this fictitious scenario. What is the color of the bean?

Enumerator instruction: Have the participant draw a bean.

Enumerator prompt if participant draws a white bean: Ok, you drew a white bean. This means that the NGO would not audit you, so you would get the following allocation

- NGO: [slider value] PKR
- You: [500 – slider value] PKR

Enumerator prompt if participant draws a black bean: Ok, you drew a black bean. This means that the NGO would audit you, so you would get

- NGO: [slider value] PKR
- You: [500 – slider value] PKR

Game A Real Round Two Now we will play another round of the activities. Remember, while we are talking about hypothetical money in the activities, so you should treat this money like it is real. We will play each of these activities twice, and for each activity, we will randomly pick one of your answers, and you will be transferred the money over JazzCash in the coming days.

You have 500 PKR, and can decide how much to allocate to yourself and how much to allocate to the government fund. The money allocated to the government fund will be given to government-affiliated institutions to be used to help with local programs in your community. Any money allocated to the government fund will be doubled.

Question. How much do you want to give to the government fund?

Enumerator instruction: Hand over the tablet so that they can pick their answer

Question. Based on your answer, this is what you and the government would get

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

Is this what you want?

Now, you have 500 PKR, and can decide how much to allocate to yourself and how much to allocate to the sports fund. Any money allocated to the sports fund will be doubled.

Question. How much do you want to give to the sports fund?

Enumerator instruction: Hand over the tablet so that they can pick their answer

Question. Based on what your answer, this is what you and the sports fund would get

- Sports Fund: [slider value] PKR
- You: [500 – slider value] PKR

Is this what you want?

Game B Real Round Two You have earned 500 PKR. Just like in real life, you must report the income you earn to the government. Also just like in real life, you must pay taxes on the income you report. These taxes are not pretend. At the end of the activity

we will give all the taxes we collect to the government. The tax rate for this activity is 25%. That means that for every 100 PKR you report, the government will take 25 PKR and you will keep 75 PKR for yourself. Also just like in real life, the government will not know how much income you earn. That means you must decide whether to report all of your income, some of your income, or none of your income. The more income you report, the more the government will take from you in taxes. Just like in real life, the government can decide to investigate you to find out if you did not report some of your income. If the government decides to investigate you, it will compare the income you earned to the income you reported. If the government finds out that you earned more than you reported, it will punish you with a fine of 50% of the income you did not report. It will also tax you on the income you reported. Also just like in real life, the government will not investigate everybody. After every round of the activity you will pick a bean from this bag. That bean will tell you if you will be investigated or not. If you pick a black bean, that means you will be investigated. If you pick a white bean, that means you will not be investigated. There are 18 white beans in the bag, but only 2 black beans. That means the chance you will be investigated is 10%, or 1 in 10.

Question. How much do you decide to report to the government?

Enumerator instruction: Hand over the smartphone/tablet to the participant, and show them how they can enter their allocation.

Question. Based on what your answer, this is what you and the government would get if you draw a white bean

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

and this is what you and the government would get if you draw a black bean

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

Is this what you want?

Ok, now that you have decided how much to allocate, let's draw one of the beans to see if the government would audit you in this fictitious scenario. What is the color of the bean? *Enumerator instruction: Have the participant draw a bean.*

Enumerator prompt if participant draws a white bean: Ok, you drew a white bean. This means that the government would not audit you, so you would get the following allocation

- Government Fund: [slider value] PKR
- You: [500 – slider value] PKR

Enumerator prompt if participant draws a black bean: Ok, you drew a black bean. This means that the government would audit you, so you would get

- Government Fund: *[slider value]* PKR
- You: *[500 – slider value]* PKR

Game B Variant Real Round Two Imagine you received funding from a local NGO to support a small business run by you. The funding works the following way: Only those people who made a profit need to pay a share to the NGO for the profits they made. This way, those who did better pay more, just like in real life. In this activity, you will decide how much profit to report to the NGO.

It turns out you made 500 PKR in profits. You now need to decide how much of these profits to report to the NGO. Remember, even though the activity is fun, it is not really a game. Similarly to the previous activity, the NGO takes a 25% share of the profits you made. That means that for every 100 PKR you report, the NGO will take 25 PKR and you will keep 75 PKR for yourself.

Also just like in real life, the NGO will not know how much profit you made. That means you must decide whether to report all of your profit, some of your profit, or none of your profit. The more profit you report, the more the NGO will take from you as its share.

Like I said before, the NGO will not know how much profit you made. But just like in real life, the NGO can decide to investigate you to find out. If the NGO decides to investigate you, it will compare the profit you reported to the profit you made. If the NGO finds out that you made more than you reported, it will punish you with a fine of 50% of what you don't report. It will also collect its share on all the profit you reported, at the usual 25% rate. Also just like in real life, the NGO will not investigate everybody. After every round of the activity you will pick a bean from this bag. That bean will tell you if you will be investigated or not. If you pick a black bean, that means you will be investigated. If you pick a white bean, that means you will not be investigated. There are 18 white beans in the bag, but only 2 black beans. That means the chance you will be investigated is 10%, or 1 in 10.

Question. How much do you decide to report to the NGO?

Enumerator instruction: Hand over the smartphone/tablet to the participant, and show them how they can enter their allocation.

Question. Based on your answer, this is what you and the NGO would get if you draw a white bean

- NGO: *[slider value]* PKR
- You: *[500 – slider value]* PKR

and this is what you and the NGO would get if you draw a black bean

- NGO: *[slider value]* PKR
- You: $500 - \textit{slider value}$ PKR

Is this what you want?

Ok, now that you have decided how much to allocate, let's draw one of the beans to see if the NGO would audit you in this fictitious scenario. What is the color of the bean?

Enumerator instruction: Have the participant draw a bean.

Enumerator prompt if participant draws a white bean: Ok, you drew a white bean. This means that the NGO would not audit you, so you would get the following allocation

- NGO: *[slider value]* PKR
- You: $500 - \textit{slider value}$ PKR

Enumerator prompt if participant draws a black bean: Ok, you drew a black bean. This means that the NGO would audit you, so you would get

- NGO: *[slider value]* PKR
- You: $500 - \textit{slider value}$ PKR

F Survey Instruments

F.1 Baseline Survey

Q#	Question Text	Possible Answers
A1	What is your marital status?	Single; Married/cohabitation; Separated; Divorced; Widowed; Refused to answer; Don't know
A2	Including yourself, how many people are there in total in your household, living and eating together in the same house?	Open response (integer)
A3	Who handles your household's financial decisions, for example how much money to save and what to buy with the household's money?	You; Your husband/partner; You AND your partner together; Some other male household member; Some other female household member; Refused to answer; Don't know
A4	Are you the head of your household?	Yes; No
B1	In the last 30 days, did the head of your household work for any organization, individual or on own's account (in a business enterprise belonging to the household or member of the household, e.g. as a trader, barber, shop owner, dressmaker, carpenter, taxi driver, etc)?	Yes worked for organisation; Yes worked for individual; Yes worked for self; No did not work; Refused to answer; Don't know
B2	You mentioned that you are NOT the head of your household. In the last 30 days, did you work for any organization, individual or on own's account (in a business enterprise belonging to the household or member of the household, e.g. as a trader, barber, shop owner, dressmaker, carpenter, taxi driver, etc)?	Yes worked for organisation; Yes worked for individual; Yes worked for self; No did not work; Refused to answer; Don't know
B3	You mentioned that you are the head of your household. In the last 30 days, did you work for any organization, individual or on own's account (in a business enterprise belonging to the household or member of the household, e.g. as a trader, barber, shop owner, dressmaker, carpenter, taxi driver, etc)?	Yes worked for organisation; Yes worked for individual; Yes worked for self; No did not work; Refused to answer; Don't know
B4	In the last 30 days, how much income did all the members of your household earn from economic activity in total? (eg. Wages/Salaries from work including profit from your business, etc)	Open response (integer)

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Q#	Question Text	Possible Answers
B5	Is your household currently a beneficiary of the Benazir Income Support Program (BISP)?	Yes; No
B6	Imagine that you have an emergency and you need to come up with 2,500 PKR. How difficult is it that you could come up with this amount within the next 30 days? Would you say it is very difficult, somewhat difficult, somewhat easy, or very easy?	Very difficult; Somewhat difficult; Somewhat easy; Very easy; Refused to answer; Don't know
B7	In the last 30 days have you personally transferred airtime to or received airtime from a relative or friend living in a different area inside Pakistan through a mobile phone?	Yes transferred airtime; Yes received airtime; No; Refused to answer; Don't know
B8	In the last 30 days, have you personally transferred money to or received money from a relative or friend living in a different area inside Pakistan through a mobile phone?	Yes transferred money; Yes received money; No; Refused to answer; Don't know
B9A	In the last 30 days, did you have any medical needs to buy medicine?	Yes; No; Refused to answer; Don't know
B9B	Were you able to pay for the medicine for these medical needs?	Yes; No; Refused to answer; Don't know
D1	Over the past 7 days, how many days did you or any other adults in your household skip meals because there were not enough resources for food?	Open response (integer)
D2	In the last 7 days, were your children ever forced to skip a meal because there wasn't enough money for food?	Yes; No; Refused to answer; Don't know
D3	Over the past 7 days, how many days did you or any household member go to sleep at night hungry because there was not enough food?	Open response (integer)
D4	Over the past 7 days, how many days did you or any household member go a whole day and night without eating anything at all because there was not enough food?	Open response (integer)
E1	How many days over the last 7 days, did most members of your household (50% +) eat the following food items?	[Note]
E2	Cereals, grains, roots and tubers	Open response (integer)
E3	Pulses/ legumes	Open response (integer)
E4	Milk and other dairy products	Open response (integer)
E5	Meat, fish and eggs	Open response (integer)

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Q#	Question Text	Possible Answers
E6	Vegetables and leaves	Open response (integer)
E7	Fruits	Open response (integer)
E8	Oil/fat/butter	Open response (integer)
E9	Sugar, or sweet	Open response (integer)
F1	How much do you agree or disagree with the following statement: I am highly satisfied with my present financial condition.	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
F2	All things considered, how satisfied are you with your life as a whole these days?	Open response (integer)
F3	Taking all things together, would you say you are:	Very happy; Quite happy; Not very happy; Not at all happy; Refused to answer; Don't know
F4	How do you compare the overall economic situation of the household with 30 days ago?	Much worse; Slightly worse; Same; Slightly better; Much better; Refused to answer; Don't know
G1	In your personal opinion, what should be the Pakistan's government most important priority to help women like you?	Addressing climate change; Improving health services; Improving education services; Reducing corruption; Providing direct income support (e.g. food or money); Improving the provision of electricity
G2	Has your household witnessed any of the following shocks in the last 12 months?	Flood, fire or other natural disaster destroys house/property/business; Theft, burglary or other man-caused loss of house/property/family land/business; Bad weather or pests destroy all or part of crops or your livestock dies due to a disease; Job Loss / Reduced income of the main wage-earner; Death / Loss of livestock; Death or disability of main income-earner; Loss of cash, including savings; Major medical emergency; Forced displacement due to violence; Nothing; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
G3	Do you agree or disagree with the following statement: The government's tax authorities always have the right to make people pay taxes?	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
H1	During the last 7 days, were there days (and, if so, how many) when your household had to rely on less preferred and less expensive food to cope with a lack of food or money to buy it?	Open response (integer)
H2	During the last 7 days, were there days (and, if so, how many) when your household had to borrow food or rely on help from a relative or friend to cope with a lack of food or money to buy it?	Open response (integer)
H3	During the last 7 days, were there days (and, if so, how many) when your household had to limit portion size of meals at meal times to cope with a lack of food or money to buy it?	Open response (integer)
H4	During the last 7 days, were there days (and, if so, how many) when your household had to restrict consumption by adults in order for small children to eat to cope with a lack of food or money to buy it?	Open response (integer)
H5	During the last 7 days, were there days (and, if so, how many) when your household had to reduce number of meals eaten in a day to cope with a lack of food or money to buy it?	Open response (integer)
H6	How many members of your household have physical or mental impairments?	Open response (integer)
H7	How many other members of your household have been sick or chronically ill for at least 3 months over the past 12 months (do not count disabled household members included above)?	Open response (integer)
I1	Did household members being chronically or acutely ill receive medical attention while sick?	Yes, some of them; No; Yes, all of them
I2	What is the main source of energy for cooking used by your household?	None; Firewood (Purchased); Firewood (Collected); Charcoal; Gas; Electricity; Animal dung; Paraffin; Kerosen; Solar Energy; Straw; Other

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Q#	Question Text	Possible Answers
I3	What is the main source of lighting used by your household?	Fire (wood, straw, etc.); Charcoal; Gas; Public electricity provider; Private electricity provider; Rural electrification plan; Candles; Generator; Simple solar lantern; Solar energy kit (several lamps); Full solar home system (sufficient for several lamps and electric appliances); Torch; Oil; Other
I4	What is the MAIN source of drinking water for your household?	Piped Water (Inside or outside the house); Public tap/standpipe; Tube well/borehole (Handpump and pump); Protected source (dug well, protected spring, rainwater collection); Unprotected source (dug well, protected spring, rainwater collection); Tanker truck; Small water vendor; Bottled water; Other
I5	Does your household currently feel safe in your environment?	Yes; No; Refused to answer; Don't know
I6	Has your household experienced any insecurity or violence in the last 12 months?	Yes; No; Refused to answer; Don't know

F.2 Follow-up Survey

Q#	Question Text	Possible Answers
A1	In the last six months, have you received any kind of assistance from someone? By assistance we mean any money, food, or donations that you didn't have to pay for.	Yes; No; Refused to answer; Don't know
A2	In the last six months, have you received any kind of assistance from someone outside this program? By assistance we mean any money, food, or donations that you didn't have to pay for.	Yes; No; Refused to answer; Don't know
A3	Who did this assistance come from?	The Pakistan national government; An organisation affiliated with the national government; Local community leaders; PPAF; JazzCash; Another Pakistani non-governmental organization; A donor from outside Pakistan; The SRSO; Other; Refused to answer; Don't know
A4	What kind of assistance did you receive?	Cash; Money through your CNIC; Money through some other mean; Food; Other in-kind type (clothes, school materials); Assets (live-stock, equipment); Other; Refused to answer; Don't know
B1A	In the last 30 days, did the head of your household work for any organization, individual or on own-account (in a business enterprise belonging to the household or member of the household, - e.g. as a trader, barber, shop owner, dressmaker, carpenter, taxi driver, etc)?	Yes worked for organisation; Yes worked for individual; Yes worked for self; No did not work; Refused to answer; Don't know
B1B	You mentioned that you are NOT the head of your household. In the last 30 days, did you work for any organization, individual or on own-account (in a business enterprise belonging to the household or member of the household, - e.g. as a trader, barber, shop owner, dressmaker, carpenter, taxi driver, etc)?	Yes worked for organisation; Yes worked for individual; Yes worked for self; No did not work; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
B1C	You mentioned that you are the head of your household. In the last 30 days, did you work for any organization, individual or on own-account (in a business enterprise belonging to the household or member of the household, - e.g. as a trader, barber, shop owner, dressmaker, carpenter, taxi driver, etc)?	Yes worked for organisation; Yes worked for individual; Yes worked for self; No did not work; Refused to answer; Don't know
B2A	In the last 30 days, how much income did all the members of your household earn from economic activity in total? (eg. Wages/Salaries from work including profit from your business, etc).	Open response (integer, [0-300000 PKR])
B2B	In the last 30 days, how much income did all the members of your household earn from economic activity in total? (eg. Wages/Salaries from work including profit from your business, etc). If you have received any JazzCash payments, please exclude it from this total.	Open response (integer, [0-300000 PKR])
B3A	Do you currently save?	Yes; No; Refused to answer; Don't know
B3B	How much did you save in the last 30 days?	Open response (integer, [0-300000 PKR])
B4	In the last 30 days, have you personally performed any income-generating activity?	Yes; No; Refused to answer; Don't know
B5A	In the last 30 days, did you have any medical needs to buy medicine?	Yes; No; Refused to answer; Don't know
B5B	Were you able to pay for the medicine for these medical needs?	Yes; No; Refused to answer; Don't know
B6	How do you compare the overall economic situation of the household with 30 days ago?	Much worse; Slightly worse; Same; Slightly better; Much better; Refused to answer; Don't know
C1	Over the past 7 days, how many days did you or any other adults in your household skip meals because there were not enough resources for food?	Open response (integer, [0-7 days])
C2	In the last 7 days, were your children ever forced to skip a meal because there wasn't enough money for food?	Yes; No; Refused to answer; Don't know
C3	Over the past 7 days, how many days did you or any household member go to sleep at night hungry because there was not enough food?	Open response (integer, [0-7 days])

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Q#	Question Text	Possible Answers
C4	Over the past 7 days, how many days did you or any household member go a whole day and night without eating anything at all because there was not enough food?	Open response (integer, [0-7 days])
C5	How many days over the last 7 days, did most members of your household (50% +) eat the following food items?	[Note: Full text verbatim]
C5A	Cereals, grains, roots and tubers	Open response (integer, [0-7 days])
C5B	Pulses/ legumes / nuts	Open response (integer, [0-7 days])
C5C	Milk and other dairy products	Open response (integer, [0-7 days])
C5D	Meat, fish and eggs	Open response (integer, [0-7 days])
C5E	Vegetables and leaves	Open response (integer, [0-7 days])
C5F	Fruits	Open response (integer, [0-7 days])
C5G	Oil/fat/butter	Open response (integer, [0-7 days])
C5H	Sugar, or sweet	Open response (integer, [0-7 days])
C6	How many meals were eaten yesterday (during the day and night) by household members in the following age categories:	[Note: Full text verbatim]
C6A	0-4 years	Open response (integer, [0-5 meals])
C6B	5-17 years	Open response (integer, [0-5 meals])
C6C	18+ years	Open response (integer, [0-5 meals])
D1	How much do you agree or disagree with the following statement: "I am highly satisfied with my present financial condition."	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
D2	All things considered, how satisfied are you with your life as a whole these days?	Open response (integer, [1-10])
D3	Taking all things together, would you say you are:	Very happy; Quite happy; Not very happy; Not at all happy; Refused to answer; Don't know
D4	How strongly do you agree with the following statement: "I feel like my life has importance"	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
D5	Generally speaking, would you say that most people can be trusted, or that you can't be too careful in dealing with people?	Most people can be trusted; You cannot be too careful in dealing with people; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
E1	In the last 30 days, do you know anyone in your community who has been approached by government officials to provide them with any kind of assistance, such as food or money?	Yes; No; Refused to answer; Don't know
E2	In the last 30 days, do you know anyone in your community who has been approached by a local community leader to provide them with any kind of assistance, such as food or money?	Yes; No; Refused to answer; Don't know
E3	In the last 30 days, have you been approached by government officials to provide them with any kind of assistance, such as food or money?	Yes; No; Refused to answer; Don't know
E4	In the last 30 days, have you been approached by a local community leader to provide them with any kind of assistance, such as food or money?	Yes; No; Refused to answer; Don't know
F1	How much do you agree with the following statement: "The work of international organisations is helping Pakistan."	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
F2	How much do you agree with the following statement: "The work of international organisations is helping citizens like me in Pakistan."	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
F3	Do you find the current national government to be effective in providing services to the average person?	Yes; No; Refused to answer; Don't know
F4	Generally speaking, do you trust the current national government?	Yes; No; Refused to answer; Don't know
F5	Generally speaking, do you trust local government officials?	Yes; No; Refused to answer; Don't know
F6	Thinking about other women like you, not yourself, how satisfied do you think they are with the current national government?	Open response (integer)
F7	For each of the following statements, please tell me whether you agree or disagree: The government's tax authorities always have the right to make people pay taxes?	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
F8	The police always have the right to make people obey the law?	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
F9	Courts have the right to make decisions that people always have to abide by?	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
F10	Are you satisfied with the current national government?	Yes; No; Refused to answer; Don't know
G1	On a scale from 0 to 10, 0 being completely unlikely and 10 being completely likely, how likely do you think it is that the political system will change a lot in the next 2 years?	Open response (integer, [0-10])
G2	In the next 2 years, do you think the political system in Pakistan will be better for people like you?	Yes; No; Refused to answer; Don't know
G3	Generally speaking, do you believe the Government of Pakistan is going in the right direction, the wrong direction, or is in the same place, not going anywhere?	Wrong direction; Same place; Right direction; Refused to answer; Don't know
H1	In the last 30 days, have you tried to influence a male family member (husband, father, brother) to raise or act on a community issue?	Yes; No; Refused to answer; Don't know
H2	In the last 30 days, have you encouraged or helped someone in your household to attend a community meeting or take action on a community issue?	Yes; No; Refused to answer; Don't know
H3	In the last 30 days, have you spoken to a local leader (such as a neighborhood elder, religious scholar, or social worker) about issues affecting your community, such as safety or access to healthcare and education?	Yes; No; Refused to answer; Don't know
H4	In the last 30 days, have you participated in or worked with a women's group, community organization, or political platform to address challenges faced by women, such as access to employment, domestic violence support, or government assistance programs?	Yes; No; Refused to answer; Don't know
H5	In the last 30 days, have you contributed resources (money, food, or labor) to support a community initiative, even if you did not directly participate?	Yes; No; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
H6	In the last 30 days, have you helped other women in your community by sharing information, organizing meetings, or encouraging them to participate in discussions or initiatives related to their rights, well-being, or economic opportunities?	Yes; No; Refused to answer; Don't know
I1	Suppose you have earned 500 PKR, but you have to give away the money to two other people. You can't keep any of the money for yourself. Assume that these two people have the same standard of living. Now suppose that one of the two people is from Pakistan, and the other person is from Afghanistan. How much of your 500 PKR would you give to the random stranger from Pakistan, if the rest goes to the random stranger in Afghanistan?	Open response (integer)
I2	Some people feel they are responsible only for their immediate family, while others feel responsible for their neighbors or friends, too. Which best describes your view?	Only my immediate family is my responsibility; My family is my main responsibility, but I should help neighbors if I can; I feel responsible for anyone in my community who needs help; Refused to answer; Don't know
I3	Some people say that if a person is strong or wealthy, they have a moral responsibility to help those who are weaker or poorer. Others say helping is a personal choice, not a responsibility. Which of the following statements comes closer to your thinking?	It's a moral duty for the strong or wealthy to help the poor; It's good to help, but it should not be called a duty; Everyone must solve their own problems; help is extra kindness, not an obligation; Refused to answer; Don't know
I4	How much do you agree or disagree with the following statement: "In Pakistan, there are many different ethnic groups (Sindhis, Baloch, Muhajirs, Brohis, etc). If one ethnic group becomes richer, this generally comes at the expense of other groups in the country."	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
I5	How much do you agree or disagree with the following statement: "In Pakistan, there are many different income classes. If one group becomes wealthier, it is usually the case that this comes at the expense of other groups."	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
I6	Some people believe we should respect all religious or cultural practices, even those different from our own. Others say we should focus on our traditions only. Which statement is closest to what you believe?	We should mainly follow our own tradition; other beliefs do not concern us; It's acceptable to learn about different beliefs, but we don't have to respect them all; All beliefs deserve equal respect, even if we don't agree with them; Refused to answer; Don't know
I7	Some say children should always obey elders and follow family rules. Others say children should have the freedom to choose their own path, even if it goes against family wishes. Which of these do you agree with most?	Children must obey their parents in all important matters; Children can share their thoughts, but parents make the final decisions; Children should have freedom to decide things like education or career; Refused to answer; Don't know
I8	Some people say everyone-women, men, rich, poor-deserves the same treatment in all aspects of life. Others say some groups deserve more respect or priority. Which comes closest to your view?	All groups deserve the same respect and opportunities; Those with higher status (e.g., wealth, family position) should get more respect; We should treat most people equally, but there are certain exceptions; Refused to answer; Don't know
J1	Who handles your household's financial decisions, for example how much money to save and what to buy with the household's money?	You; Your husband/partner; You AND your partner together; Some other male household member; Some other female household member; Refused to answer; Don't know
J2	Did you tell your partner or other household members you were receiving these payments?	Yes; No; Refused to answer; Don't know
J3	Did receiving the humanitarian aid payment in the last weeks lead to any problems between you and other household members?	Yes; No; Refused to answer; Don't know
J4	If a wife has earned some money, does she have the right to decide how to spend it without asking the permission of her husband?	Yes; No; Refused to answer; Don't know
J5	In the last 30 days, did you and your spouse or other male household members argue about managing money?	Yes; No; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
J6	To what degree do you agree with the following statement: A husband and wife can share power	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
J7	To what degree do you agree with the following statement: Women's opinions are valuable and should always be considered when household decisions are made	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
J8	Do you agree that a wife has the right to express her opinion when she disagrees with what her husband is saying?	Yes; No; Refused to answer; Don't know
J9	Are you currently able to leave the house to complete day-to-day tasks like buying groceries and medicine?	Yes; No; Refused to answer; Don't know
J10	Are you currently able to leave the house to complete these day-to-day tasks by yourself?	Yes; No; Refused to answer; Don't know
K1	There is an NGO in your local area that focuses on gender issues. They are writing a petition asking local government officials for more rights and liberties for women like you in your area. Would you be willing to sign the petition? If you want to sign the petition, we will give your name to the NGO, and your name will appear in the petition that will be made publicly available. You won't need to appear anywhere, or do anything other than say whether you want to appear in the petition or not. If you don't want to sign the petition, that is also alright and there would be no issues with that.	Yes; No; Refused to answer; Don't know
K2	In February, you received one humanitarian aid payment worth 7000 PKR through JazzCash. Who do you think donated the money that was sent to you?	The Pakistan national government; An organisation affiliated with the national government; Local community leaders; PPAF; JazzCash; Another Pakistani non-governmental organization; A donor from outside Pakistan; The SRSO; Other; I don't know; Refused to answer

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Q#	Question Text	Possible Answers
K3	In February, you participated in an event related to a humanitarian assistance program. Who do you think donated the money that financed that humanitarian assistance program?	The Pakistan national government; An organisation affiliated with the national government; Local community leaders; PPAF; JazzCash; Another Pakistani non-governmental organization; A donor from outside Pakistan; The SRSO; Other; I don't know; Refused to answer
K6	How confident are you that XXX paid for your humanitarian aid payment?	Open response (integer, [0-10])
K7	How confident are you that XXX paid for the humanitarian assistance program?	Open response (integer, [0-10])
L1	I'd like now to play a game called the Digits Forward game. It's designed to measure your short term memory and is not a test of intelligence. Listen carefully as I say some numbers. When I finish, you should repeat them back to me in the same order I said them to you. For example: if I say 7866, you say 7866. Or if I say 1112, you say? (Answer: 1112. Even if the student does not answer correctly, you will continue with the test.).	[Note]
L2	I will read each set of numbers only once so be careful to pay attention the first time. If you miss what I've said the first time, just try your best to repeat whatever you heard.	[Note]
L3	We will start with a three digit number	[Note]
L3A	3 - 8 - 6	Yes; No; Refused to answer; Don't know
L3B	6 - 1 - 2	Yes; No; Refused to answer; Don't know
L4	We will now do a four digit number	[Note]
L4A	3 - 4 - 1 - 7	Yes; No; Refused to answer; Don't know
L4B	6 - 1 - 5 - 8	Yes; No; Refused to answer; Don't know
L5	We will now do a five digit number	[Note]
L5A	8 - 4 - 2 - 3 - 9	Yes; No; Refused to answer; Don't know
L5B	5 - 2 - 1 - 8 - 6	Yes; No; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
L6	We will now do a six digit number	[Note]
L6A	3 - 8 - 9 - 1 - 7 - 4	Yes; No; Refused to answer; Don't know
L6B	7 - 9 - 6 - 4 - 8 - 3	Yes; No; Refused to answer; Don't know
L7	We will now do a seven digit number	[Note]
L7A	5 - 1 - 7 - 4 - 2 - 3 - 8	Yes; No; Refused to answer; Don't know
L7B	9 - 8 - 5 - 2 - 1 - 6 - 3	Yes; No; Refused to answer; Don't know
M1	You should have received payment for PKR 7000 as part of your participation in this study. Have you spent at least some of the PKR 7000 you received?	Yes; No; Refused to answer; Don't know
M2	Why haven't you spent any of the 7000 PKR you received?	I was not aware of this payment; I am saving the funds for when I need them; I do not know how to access the funds; I do not know how to spend the funds; I haven't had time to spend the funds; Other; Refused to answer; Don't know
M3	(Aid Usage Section Part 1) Did you experience any difficulty receiving this payment via the JazzCash app?	Yes; No; Refused to answer; Don't know
M4	(Aid Usage Section Part 1) Did anybody approach you for part of that payment?	No; Yes, my husband/partner; Yes, another male household member; Yes, another female household member; Yes, a government official; Yes, other; Refused to answer; Don't know
M5	(Aid Usage Section Part 1) Did anyone else decide how to spend your direct aid payment?	No; Yes, my husband/partner; Yes, another male household member; Yes, another female household member; Yes, a government official; Yes, other; Refused to answer; Don't know
M6	From the 7000 PKR payment you received, have you used all or part of it for the following:	[Note]
M7	To buy directly from merchants using JazzCash	Yes; No; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
M8	To transfer money to someone else inside the household	Yes; No; Refused to answer; Don't know
M9	To transfer money to someone else outside the household	Yes; No; Refused to answer; Don't know
M10	To buy airtime	Yes; No; Refused to answer; Don't know
M11	To pay for bills	Yes; No; Refused to answer; Don't know
M12	To buy food	Yes; No; Refused to answer; Don't know
M13	To buy clothes	Yes; No; Refused to answer; Don't know
M14	To buy medicine	Yes; No; Refused to answer; Don't know
M15	To invest in a business	Yes; No; Refused to answer; Don't know
M16	To pay for transport	Yes; No; Refused to answer; Don't know
M17	To pay rent	Yes; No; Refused to answer; Don't know
M18	To pay debt	Yes; No; Refused to answer; Don't know
M19	To save	Yes; No; Refused to answer; Don't know
M20	Do you share your JazzCash account with someone else?	Yes; No; Refused to answer; Don't know
M21	Who do you share your JazzCash account with?	Husband/partner; Parents; Sons; Other household member; Someone outside your household; Other; Refused to answer; Don't know
N1	Which of the following have you done with your funds?	I have bought goods directly from merchants using my JazzCash wallet; I have cashed out money; I have transferred money to others; Other; It's in the wallet; Refused to answer; Don't know
N2	When you bought from merchants using your JazzCash wallet, did you experience any problems?	Yes; No; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
N3	(Cash vs. Digital Section Part 1) When you cashed out your funds, did you cash out all your funds, or just a part?	Cashed out all the funds; Cashed out only part of the funds; Refused to answer; Don't know
N4	(Cash vs. Digital Section Part 1) From the money you have cashed out, did you cash out everything at once, or did you cash out on different occasions?	Cashed out at once; Cashed out in different occasions; Refused to answer; Don't know
N5	(Cash vs. Digital Section Part 1) Usually when you cash out, for how long do you need to travel to cash out your funds?	Open response (integer, in minutes)
N6	(Cash vs. Digital Section Part 1) Usually when you cash out, how much do you spend on transportation to cash out your funds?	Open response (integer, in PKR)
N7	(Cash vs. Digital Section Part 1) Usually when you cash out, for how long do you need to wait in line to cash out your funds?	Open response (integer, in minutes)
N8	(Cash vs. Digital Section Part 1) When you have cashed out, did you experience any problems?	Yes; No; Refused to answer; Don't know
O1	Please tell me, in general, how willing or unwilling you are to take risks, using a scale from 0 to 10, where 0 means you are "completely unwilling to take risks" and 10 means "very willing to take risks." You can also use any number between 0 and 10 to indicate where you fall on the scale.	Open response (integer, [0-10])
O2	How well does each of the following statements describe you as a person? Please indicate your answer on a scale from 0 to 10. A 0 means "does not describe me at all," and a 10 means "describes me perfectly." You can also use any number between 0 and 10 to indicate where you fall on the scale.	[Note]
O3	When someone does me a favor, I am willing to return it.	Open response (integer)
O4	If I am treated very unjustly, I will take revenge at the first occasion, even if there is a cost to do so.	Open response (integer)
O5	I assume that people have only the best intentions.	Open response (integer)
O6	I feel comfortable speaking in front of a small group.	Open response (integer)

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Q#	Question Text	Possible Answers
O7	We now ask you for your willingness to act in a certain way. Please again indicate your answer on a scale from 0 to 10, where 0 means “completely unwilling to do so,” and a 10 means “very willing to do so.” You can also use any number between 0 and 10 to indicate where you fall on the scale.	[Note]
O8	How willing are you to punish someone who treats you unfairly, even if there may be costs for you?	Open response (integer)
O9	How willing are you to punish someone who treats others unfairly, even if there may be costs for you?	Open response (integer)
O10	How willing are you to give to good causes without expecting anything in return?	Open response (integer)
P1	Have you interacted with government officials as part of their official job in the past 30 days?	Yes; No; Refused to answer; Don’t know
P2	How interested would you say you are in regards to topics related to the government?	Open response (integer, [0-10])
P3	What are the most important sources of information for you to learn about what the government is doing? [Do not prompt, list up to three]	Relatives, friends, neighbors; Community bulletin board; Local market; Community or local newspaper; National newspaper; Radio; Television; Groups or associations; Business or work associates; Political associates; Community leaders; Government agents; NGOs; Internet; Other; Does not get news about what the government is doing; Refused to answer; Don’t know
P4	Did you vote in the general election in February last year?	Yes; No; Refused to answer; Don’t know
P5	In the last 10 years, did you vote in national or local elections?	Yes; No; Refused to answer; Don’t know
P6	Do you plan to vote in the next election?	Yes; No; Refused to answer; Don’t know
P7	Have you ever belonged to political groups or organisations?	Yes; No; Refused to answer; Don’t know
P8	When you get together with your friends, how frequently do you discuss government matters?	Never; Rarely; Sometimes; Frequently; Always; Refused to answer; Don’t know

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Q#	Question Text	Possible Answers
P9	People may hold critical attitudes toward the government. If people in Pakistan hold critical attitudes toward the government, to what extent do you think they would be afraid of expressing their true attitudes in public?	Not at all afraid; A little afraid; Somewhat afraid; Very afraid; Extremely afraid; Refused to answer; Don't know
P10	Women in Pakistan participate much less in politics than men, in terms of running in elections, voting, participating in political movements and so on. What do you think is the main reason for this?	Women feel left out of politics; Women don't care about politics; Women don't know about politics; Women should not participate in politics; Women do not have decision-making power regarding their participation in politics; Men in households act as gatekeepers, restricting women's political participation; Women's mobility is restricted due to a lack of male accompaniment, making it difficult for them to vote or campaign; Women fear harassment or social stigma for participating in politics; Cultural norms discourage women's involvement in politics; Women are too busy with household responsibilities; Political parties do not actively engage or support female candidates and voters; Other; Refused to answer; Don't know
P11	How much do you agree with the following statement: My husband or other male members of my household play an important role in my political decision-making, for example whether and for whom to vote.	Agree a lot; Somewhat agree; Neither agree nor disagree; Somewhat disagree; Not agree at all; Refused to answer; Don't know
P12	Imagine you have a minor problem with someone from your neighborhood (for example, about the boundary of your home or noise). Who would you go to first for help in settling it?	A male member of the household; A female member of the household; A local community leader; The police; A government official; A friend; Other; Refused to answer; Don't know

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Q#	Question Text	Possible Answers
P13	Imagine you have a problem about something happening in your community (for example, about a lack of health services, safety issues in your community, or problems with electricity provision). Who would you go to raise this issue?	A male member of the household; A female member of the household; A local community leader; The police; A government official; A friend; Other; Refused to answer; Don't know
P14	Do you know any household that receives aid payments as part of this program?	Yes; No; Refused to answer; Don't know
P15	Did you receive any kind of assistance (e.g. food or money) from any of the other households you know receive aid payments as part of this program?	Yes; No; Refused to answer; Don't know